



Module Code & Module Title
CU6051NP Artificial Intelligence

75% Individual Coursework

Submission: Final Submission

Academic Semester: Autumn Semester 2025

Credit: 15 credit semester long module

Student Name: Muna Pun

London Met ID: 23048838

College ID: NP04CP4A230233

Assignment Due Date: 21/01/2026

Assignment Submission Date: 21/01/2026

Submitted To: Suraj Shrestha

GitHub Link	https://github.com/Munapoon/23048838-Muna-Pun-AI
--------------------	---

I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Acknowledgements

I would like to express my heartfelt gratitude to everyone who has helped me during the process of this project on COVID-19 Prediction using Machine Learning. Most importantly, I would like to thank my project supervisor and mentor, Suraj Shrestha, who provided invaluable guidance and encouragement, as well as constructive feedback. This project has been influenced by his professional guidance, which has helped me to overcome obstacles and perfect my strategy. His commitment to teaching and mentorship is a great source of inspiration to me, and it has reinforced my desire to work in the field of data science and machine learning.

I would also like to thank my peers and colleagues who gave ideas, shared resources, read my work, and gave much-needed feedback when this project was being developed. Their contribution was critical in enhancing the quality of this work. Finally, I would like to thank Jeevan Prakash sir, who has supplied the datasets in lecturer which is used in this project. Their free access to COVID-19 data allowed me to do a significant analysis and build a model with practical use in real-life. They contributed to this project in such a way that it became not only possible but also relevant to global efforts to understand and predict the trends of pandemics.

Abstract

COVID-19 pandemic provided a serious health threat to the entire world, and ensuring the correct prediction of cases is the key to successful planning of the general health of the population. This project used supervised machine learning to forecast the trends in global COVID-19 case prevalence using historic epidemiological information. The models developed and tested were Linear Regression and Random Forest Regression models based on the publicly available datasets (worldwide-aggregate.csv and countries-aggregated.csv).

To enhance predictive performance, time-series feature engineering, such as lag values, rolling averages, and growth rates was used. The evaluation of the model was by considering the MAE, RMSE, and MAPE. The findings showed that Linear Regression had better accuracy, which suggests that there is a high linear correlation between the development of COVID-19 cases. This study proves the usefulness of properly chosen machine learning models in assisting data-driven decision-making in case of a global health emergency.

Table of Contents

1. Introduction.....	1
1.1 Introduction to AI	1
1.2 AI Concepts Used	3
1.3 Problem domain of the Topic.....	5
1.4 Aims	7
1.5 Objectives	7
1.6 Dataset Description.....	8
2. Background.	11
2.1 Research Work on the Chosen Topic.....	12
2.2 Review and Analysis of Existing Work	21
3. Solution	28
3.1 Proposed Solution / used AI algorithm	28
3.2. Pseudocode	38
3.3. Diagrammatical representations of the solution	39
3.4. Development process and Tools	43
3.5. Development and Achieved results.....	50
4. Conclusion.....	83
4.1 Work done Analysis	83
4.2 Real-World Impact and Significance	85
4.3 Further Work	86
5. References	87

Table of Figure

Figure 1:Introduction to Artificial Intelligence	1
Figure 2: Dataset figure	9
Figure 3:Research work 1	12
Figure 4:Research work done 2	14
Figure 5:Research work done 3	15
Figure 6:Research work 4	17
Figure 7: Research work 5	19
Figure 8: Existing work 1	22
Figure 9:Existing work 2	23
Figure 10: Exisiting Work 3.....	24
Figure 11:Existing work 4	25
Figure 12:Numerical of Linear	29
Figure 13: Mean Squared Error.....	30
Figure 14:Conceptual figure 1	30
Figure 15:Formula 1	32
Figure 16: Formula Description 1	33
Figure 17:Conceptual Figure 2	33
Figure 18:Lag Features	35
Figure 19: Rolling Average	35
Figure 20:Growth Rate	35
Figure 21:Time-Series Feature-Based Regression formula	36
Figure 22: Conceptual Figure 3	36
Figure 23:Flowchart of the project.....	40
Figure 24:State transition diagrams.....	41
Figure 25:Anaconda	43
Figure 26:Jupyter notebook.....	44
Figure 27:Pandas	45
Figure 28:Numpy	46
Figure 29:Scikit-Learn	46
Figure 30:Matplotlib and Seaborn	48
Figure 31:Python	48
Figure 32: The Libraries that are to be imported to predict the COVID-19.	50
Figure 33:Loaded Datasets	51
Figure 34:Output of load datasets	52
Figure 35:Data overview.....	53
Figure 36:Dataset Overview Output	54
Figure 37: Aggregate Country Data.....	55
Figure 38:Aggregate data output.....	55
Figure 39: Missing value.....	57

Figure 40: Missing value output.....	57
Figure 41:Missing value heatmap.....	58
Figure 42:Missing Values Heatmap.....	59
Figure 43:data cleaning steps	59
Figure 44: Data size after cleaning.....	60
Figure 45:Feature Engineering.....	60
Figure 46:Feature engineering output	61
Figure 47:Feature distribution.....	61
Figure 48:Feature distribution histograms	62
Figure 49:Correlation heatmap.....	63
Figure 50: Result of correlation heatmap.....	64
Figure 51:Selection of the features and train-test split	65
Figure 52: Result of feature selection	66
Figure 53:Model training and performance evaluation	66
Figure 54:Result of model comparison results	67
Figure 55:Accuracy and training time comparison.....	67
Figure 56:MAE and RMSE comparison.....	68
Figure 57: MAPE and training time comparison	69
Figure 58: Actual vs Prediction plotting code.....	70
Figure 59:Actual vs Predicted COVID-19 Confirmed Cases: Linear Regression vs Random Forest	72
Figure 60:Best model selection using low RMSE criterion	73
Figure 61:Result of best model.....	73
Figure 62:Feature importance	74
Figure 63:Feature importance Results	75
Figure 64:Final Prediction.....	75
Figure 65:Output of Final prediction	76
Figure 66: Top 10 predicted of COVID-19 Case	76
Figure 67:Visualization Output	77
Figure 68:Future prediction of COVID-19 using user input.....	79
Figure 69:Output of User data input	80

Table of Table

Table 1:Data description	10
Table 2:Model Selection and justification Table	82

1. Introduction

1.1 Introduction to AI

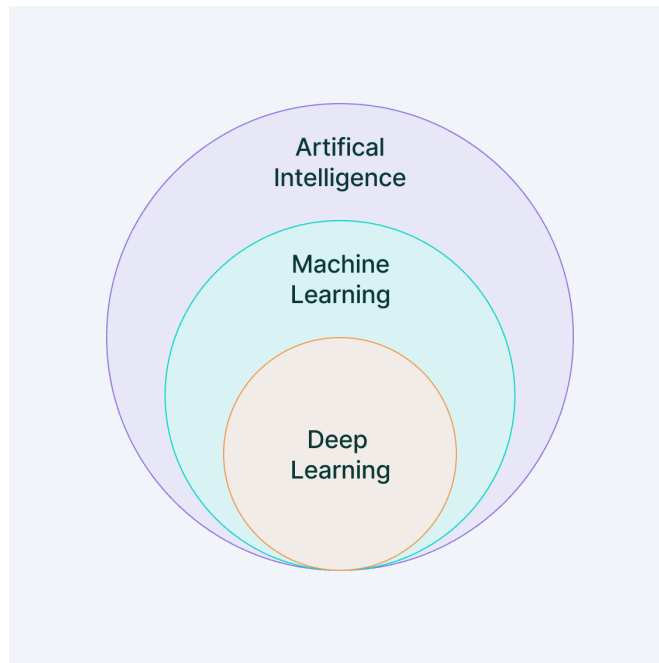


Figure 1: Introduction to Artificial Intelligence

Artificial Intelligence (AI) refers to the creation of computer systems that can detect their surroundings, process information, and respond in an intelligent way to accomplish certain goals. The AI systems are aimed at imitating human mental processes including learning, reasoning, problem-solving, and making decisions. One of the fundamental elements of AI is Machine Learning (ML), which helps systems to learn the patterns of past data and make predictions or choices without any programming (Russell, 2021).

AI and machine learning have become especially relevant in the field of healthcare, especially in reaction to worldwide health crises like the COVID-19 pandemic, in recent years. The spread of the COVID-19 to all parts of the world was at an unprecedented pace and this resulted in tremendous strain on healthcare, governments, and societies. Precise forecasting of the trends of infections became a necessity in order to be prepared with the medical needs, planning the hospital capacities, vaccination, and also in making decisions regarding policy. Conventional statistical models do not always respond to the

dynamics of changing pandemics, but machine learning models are capable of learning directly on the basis of the changing trends in data (Shinde et al., 2020).

Machine learning is important in the prediction of COVID-19 due to its ability to use mass data in the past of epidemiological activity and determine the sophisticated interactions between historical and future trends in cases. As an illustration, AI systems will predict the number of future confirmed cases, recoveries and deaths using past data of the daily number of cases. Time-series data can be used to identify growth patterns, surges, and decreases in infection rates with the help of ML algorithms, which facilitate proactive measures to ensure the well-being of the population (Tuli, 2020)

Artificial intelligence systems are capable of computing massive amounts of data, identifying trends over time, and updating to new information when it is received. The machine learning methods allow the models to keep on improving as more COVID-19 data is presented. Consequently, AI predictions have emerged as useful aids to deal with pandemics and aid evidence-based choices in the field of public health

1.2 AI Concepts Used

The machine learning concepts employed in the project in order to construct predictive models to forecast the COVID-19 cases include:

❖ **Supervised Learning:**

This project uses the supervised learning method, which involves the training of machine learning models with labelled historical data on COVID-19. Known outputs in the forms of confirmed cases, recoveries and deaths on date in the datasets used. This enables the model to be trained on the relationship between input variables (time, past case counts, growth rates and rolling averages) and the output variables (number of cases in the future). The algorithm finds hidden regularities over time to distinguish between periods of fast transmission and stable or falling ones through supervised learning. The use of supervised learning is especially appropriate to predict COVID-19 due to the presence of reliable historical data with known results that allow creating the specific prediction systems that are accurate and reliable (Russell, 2021).

❖ **Regression Algorithms:**

The regression algorithm-based supervised learning techniques are applied in the project to forecast continuous numerical values like the number of future cases of COVID-19. Regression is unlike classification problems as it looks at estimating a numerical outcome with time. The daily case counts of COVID-19 are continuous hence regression models are suitable in this problem.

Linear Regression and the Random Forest Regression are used in this project. Linear Regression offers a fundamental model to provide general trends in case growth, whereas the Random Forest Regression improves the accuracy of the prediction by offering nonlinear trends and being able to deal with noisy data. These are models that are trained on predicted time-series data with labelled data

and produce highly reliable future predictions. Past research establishes that machine learning models that are based on regression can be useful in COVID-19 forecasting activities because they are flexible and predictive (Breiman, 2001).

1.3 Problem domain of the Topic

Global Context

Since it was announced at the end of 2019, the COVID-19 pandemic was one of the most dramatic public health crises in the 21st century globally. The disease was very contagious and spread quickly within continents causing recurring outbreaks of the disease, killing many people and putting a lasting strain on the health care systems around the world. The pandemic was unpredictable and constantly changing, posing a major problem in its planning by governments and international health organizations to quickly implement the interventions. Timely global predictions of COVID-19 were of the utmost importance to predict future outbreaks, control international travel, organize the development and distribution of vaccines, and global preparedness to new variants. Consequently, prediction models that run on data have been instrumental in aiding in making evidence-based decisions in the case of the pandemic.

Local Context (Nepal)

In Nepal, the COVID-19 pandemic has revealed the existence of a number of structural issues in the national healthcare system, such as the lack of hospital facilities, medical staff, and a restricted availability of testing and critical care-facilities and resources, especially in rural and mountainous areas. The nation witnessed numerous infection waves of varying levels in various provinces and this complicated management of the whole country. Proper local forecasting was thus required to help the Government of Nepal and health authorities to plan the capacity of hospitals beds, oxygen supply, quarantine and vaccination campaigns. The use of predictive modeling with country-specific daily case information makes it possible to detect the emergence of the increasing trends at the initial stage, so the authorities can take timely containment measures in the form of lockdowns, mobility limitation, and public awareness campaigns. This localized prediction would particularly be useful in areas with limited resources such as Nepal where a proactive forecasting can help cut down on the health system workload.

AI and machine learning (ML) role

The role of artificial intelligence and machine learning in the constraint of COVID-19 forecasting has been extremely essential as it allows analyzing vast amounts of historical data to determine patterns and trends hidden in the vast amount of data. The time-series epidemiological data are best suited to supervised learning techniques due to the ability to capture time-related characteristics of lag values, rolling averages, and growth rates, especially through regression-based models. Compared to the old-fashioned compartmental models which rely on preset assumptions regarding the dynamics of the disease transmission, machine learning models can adjust under the influence of dynamic changes, which may be policy interventions, the overall behavior of the population, changes in the virus mutation, and vaccination rates. In the case of Nepal, where pandemic processes may change quickly, depending on both external and internal factors, AI-based prediction models are versatile and powerful instruments in the short-term prediction and long-term planning. These models facilitate information-based public health policies, which can assist decision-makers to react more to current and anticipated health crises.

1.4 Aims

To build a predictive model which will help to analyze the performance of the trend of COVID-19 cases across the globe and predict future tendencies in epidemics to improve public health planning.

1.5 Objectives

- To explore further in supervised learning methods specifically regression which assists in time series predictive analysis.
- To search and process the past performance information of the COVID-19 pandemic with the assistance of the time-related features which exist in the dataset.
- To combine and compare several regression models that are: Linear Regression, Random Forest Regression.
- To determine the accuracy of each model using the following metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE)

1.6 Dataset Description

The Worldwide Aggregate, Countries Aggregated datasets consist of COVID-19 vital information. Worldwide -Aggregate gives worldwide confirmed cases, recoveries, and deaths over time and the rate at which new cases are increasing. Countries -Aggregated presents statistics in each country as an example on how various regions are impacted. These data sets combined can guide us to comprehend the spread of COVID-19 in the world and compare the ways in which various countries are responding to the crisis. They pay attention to the confirmed cases, recoveries, and deaths numbers, and are critical to comprehending the development of the pandemic on a global scale.

Worldwide-Aggregate Dataset

The Worldwide Aggregate dataset has aggregated statistics of COVID-19 cases around the world and on various dates. It monitors the number of confirmed cases, healings and deaths across the globe.

Key Features:

- **Date:** This is the date on which the data was taken.
- **Confirmed:** This is the total number of confirmed COVID-19 cases that were recorded in the world on that particular date.
- **Recovered:** The overall number of people who have recovered due to COVID-19 across the whole world.
- **Deaths:** Summative number of deaths due to COVID-19 across the world.
- **Increase Rate:** This is a calculated value that demonstrates the percentage change in confirmed cases of the past day, which provides a perspective on the rate at which the virus is spreading.

Countries-Aggregated Dataset

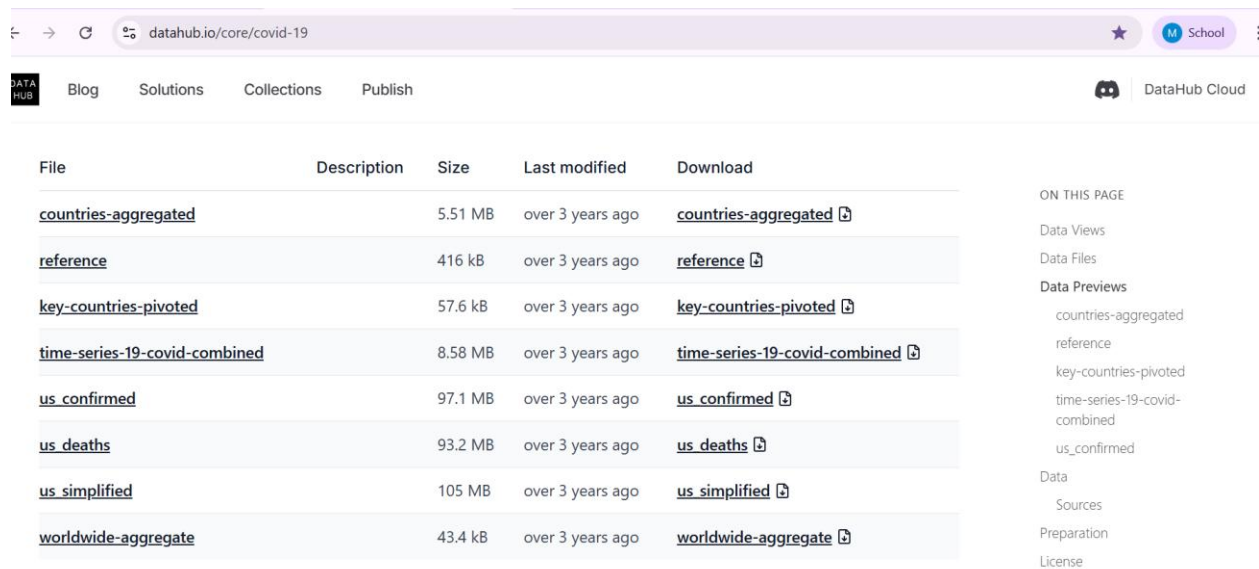
Countries Aggregated data monitors the daily statistics of COVID-19 per country and provides a more in-depth account of the impact on the country.

Key Features:

- **Date:** This is the date on which the data was taken.
- **Country:** The country to which the data is related.
- **Confirmed:** The amount of confirmed COVID-19 cases of that country on that day.
- **Recovered:** This is the total number of COVID-19 recoveries in that country that day. **Deaths:** The number of deaths in that country due to COVID-19 that day.

Datasets is chosen from here:

URL : <https://datahub.io/core/covid-19>



The screenshot shows the DataHub website interface. The browser address bar displays 'datahub.io/core/covid-19'. The website header includes the DataHub logo, navigation links (Blog, Solutions, Collections, Publish), and a 'DataHub Cloud' button. The main content area features a table with columns: File, Description, Size, Last modified, and Download. The table lists eight datasets, all of which were last modified 'over 3 years ago'. To the right of the table, a sidebar titled 'ON THIS PAGE' lists various options: Data Views, Data Files, Data Previews (with sub-links for each dataset), Data, Sources, Preparation, and License.

File	Description	Size	Last modified	Download
countries-aggregated		5.51 MB	over 3 years ago	countries-aggregated
reference		416 kB	over 3 years ago	reference
key-countries-pivoted		57.6 kB	over 3 years ago	key-countries-pivoted
time-series-19-covid-combined		8.58 MB	over 3 years ago	time-series-19-covid-combined
us_confirmed		97.1 MB	over 3 years ago	us_confirmed
us_deaths		93.2 MB	over 3 years ago	us_deaths
us_simplified		105 MB	over 3 years ago	us_simplified
worldwide-aggregate		43.4 kB	over 3 years ago	worldwide-aggregate

Figure 2: Dataset figure

Table 1:Data description

Column Name	Description	Data Type
Date	This describe on which date data was recorded.	Object
Country	Name of country for which data is recorded.	Object
Confirmed	Total number of covid cases confirmed in country on given data.	Int64
Deaths	Total number of deaths due to covid cases on given data	Int64
Recovered	Total number of covid cases recovered in country on given data.	Int64

2. Background.

Supervised learning is an important aspect of COVID-19 forecasting as it allows predicting the number of cases, recoveries, and deaths in the future using previous data. Because the COVID-19 pandemic is a continuous variable problem, such as the number of cases confirmed per day and death rates, regression models, which are a subdivision of supervised learning, are particularly appropriate in this task. These models are based on learning the relationships between past trends and patterns in time series data, and they include lag values, rolling averages, and growth rates. This method can be used to model the nonlinear dynamics of factors including viral mutations, governmental health actions, and vaccination efforts, which can be difficult to adequately model using traditional epidemiological models.

Supervised learning is also applicable in COVID-19 to inform health authorities and policymakers in making decisions under uncertainty. These models help in the allocation of resources, hospital preparedness, and communication of health to the people by coming up with dependable short-term predictions. Linear Regression techniques are used to make predictions of the base trend, whereas more sophisticated ensemble techniques, such as Random Forest, are used to make predictions with higher accuracy by including more complex interdependencies in the data. Furthermore, supervised learning systems are used to analyze scenarios by adjusting to new data in a short period of time, which makes them invaluable instruments in dealing with the dynamic COVID-19 crisis.

2.1 Research Work on the Chosen Topic

ARIMA COVID-19 Forecasting

url: <https://doi.org/10.1016/j.dib.2020.105340>

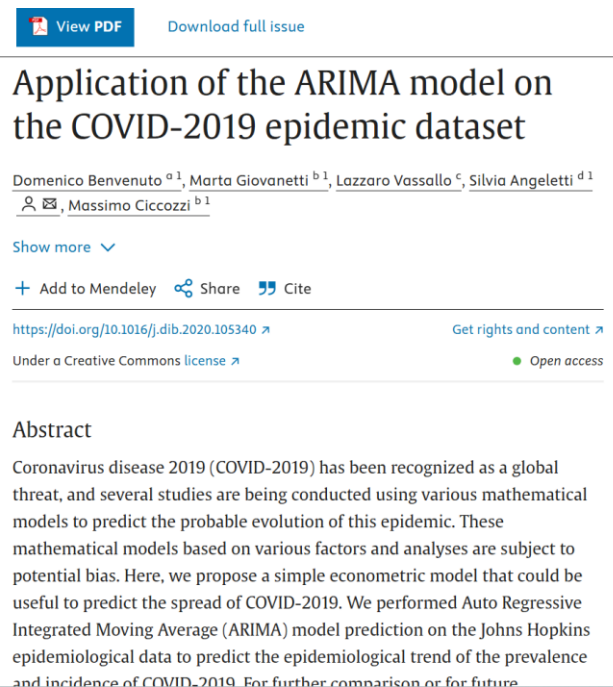


Figure 3: Research work 1

This paper concentrates on ARIMA and other time-series models that can predict COVID-19 in the short term. Although my project is not directly based on ARIMA, the study confirms the application of daily time-series data the same type of data available in the file worldwide-aggregate.csv and countries-aggregated.csv. It shows that past data can be used to learn trends in a pandemic, which is in line with my use of time-series aspects of lag values, rolling averages and growth rates in supervised learning models (Domenico Benvenuto, April 2020) .

Analysis:

The article by Domenico Benvenuto and colleagues is beneficial both to COVID-19 prediction and this project because it validates that daily historical data on cases can be successfully used to learn and predict the trend of the pandemic. Though the paper implements the ARIMA instead of the supervised learning, COVID-19 is characterized by

noticeable time-depending dynamics where the accumulated values of the past affect future values. This directly helps to use datasets like worldwide-aggregate.csv and countries-aggregated.csv in this project and the decision to include lag values, rolling averages, and growth rates as features. The fact that the short-term forecasting is better when the recent temporal tendencies are considered proves the validity of the project idea of introducing the time-series features into the supervised learning models to predict the COVID-19 more effectively.

 Random Forest & ML Models for COVID-19 Prediction

url : <https://www.sciencedirect.com/science/article/pii/S254266052030055X>

Predicting the growth and trend of COVID-19 pandemic using machine learning and cloud computing ☆

Shreshth Tuli ^a  , Shikhar Tuli ^b  , Rakesh Tuli ^c  , Sukhpal Singh Gill ^d  

Show more ▾

+ Add to Mendeley  Share  Cite

<https://doi.org/10.1016/j.iot.2020.100222> ↗

[Get rights and content](#) ↗

● Complimentary access

- Propose a novel scheme to predict the impact of COVID-19 Pandemic.
- Design a model based on Cloud Computing and Machine Learning for real-time prediction.
- Show improved prediction accuracy compared to baseline method.

Figure 4: Research work done 2

Linear Regression is a fundamental explanation given by (Shreshth Tuli, 2020) as a common baseline model used in forecasting studies. The book describes why linear models are applicable in explaining relationships among variables when real world data are complicated. This justifies the strategy of my project to apply Linear Regression to determine overall trends in the growth of COVID-19 cases prior to employing more sophisticated models, enabling me to assess the performance enhancement of the Random Forest Regression.

Analysis

The discussion presented by Shreshth Tuli (2020) justifies COVID-19 prediction and this project by emphasizing on the reliability of Linear Regression as a baseline model of predicting general trends on complex data at the real world. Linear Regression assists in recreating the overall trend of case increase over time to use as an interpretable point of reference in the context of COVID-19 forecasting. This is a direct advantage as it will

enable to assess the trends of the pandemic early and create a level of performance, which will be compared to the improvement of other parts of the model, like the Random Forest Regression, which, in turn, is more advanced.

Johns Hopkins University COVID-19 Data

url : https://coronavirus.jhu.edu/data?utm_source=chatgpt.com

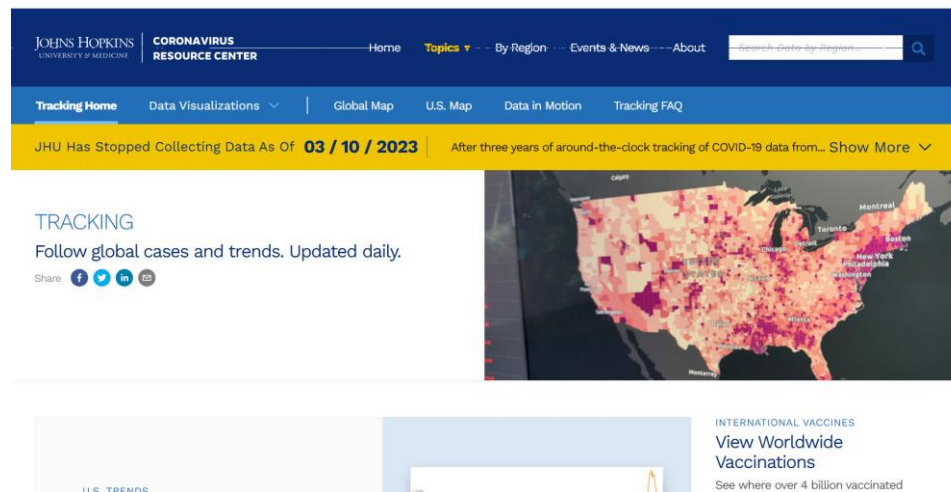


Figure 5: Research work done 3

Johns Hopkins University (JHU) COVID-19 Data Repository is directly applicable since it is a popular, constantly updated source of worldwide COVID-19 statistics. It has a similar structure, recording numbers of confirmed cases, recoveries, and deaths daily in various countries, which is similar to the worldwide-aggregate.csv and countries-aggregated.csv data used in this project. There is a large number of forecasting studies that use JHU data to test statistical and machine-learning models, thus, justifying the methodological approach taken in this case, which is supervised learning and regression techniques. The fact that the repository is regularly updated also makes it clear that it is necessary to use models that would adjust to changing data trends and indicate that the forecasting strategies in this project might be applied to real-time situations of monitoring the population health and making decisions (HopkinsUniversity, 2020).

Analysis

The Johns Hopkins University (JHU) COVID-19 Data Repository (2020), in turn, assists in COVID-19 prediction and this project in the form of a credible, real-life benchmark dataset, which confirms the predictive use of daily aggregated cases data. This structure being similar to the worldwide-aggregate.csv and countries-aggregated.csv files utilized in this project affirms that the supervised learning and regression models will be suitable in the modeling of COVID-19 trends. The extensive applicability of JHU data to the forecasting research also reflects the methodology, and its dynamic nature indicates the timeliness of creating models that would be able to adjust to the evolving nature of the pandemic and be implemented in real-time monitoring of the mass population and decision-making processes in the field of health.

 COVID-19 Global Data & Dashboard

url: <https://www.sciencedirect.com/science/article/pii/S1473309920301201>

An interactive web-based dashboard to track COVID-19 in real time

Ensheng Dong ^a, Hongru Du ^a, Lauren Gardner ^a 

Show more 

 Add to Mendeley  Share  Cite

[https://doi.org/10.1016/S1473-3099\(20\)30120-1](https://doi.org/10.1016/S1473-3099(20)30120-1) 

[Get rights and content !\[\]\(d415b5172fecdbaea44b7ff6524f4d79_img.jpg\)](#)

 Complimentary access

Referred to by

An eventful 20 years

The Lancet Infectious Diseases, Volume 21, Issue 8, August 2021, Pages 1051

 [View PDF](#)

SeroTracker: a global SARS-CoV-2 seroprevalence dashboard

The Lancet Infectious Diseases, Volume 21, Issue 4, April 2021, Pages e75-e76

Rahul K Arora, Abel Joseph, Jordan Van Wyk, Simona Rocco, Austin Atmaja, Ewan May, Tingting Yan, Niklas Bobrovitz, Jonathan Chevrier, Matthew P Cheng, Tyler Williamson, David L Buckeridge

 [View PDF](#)

Correction to *Lancet Infect Dis* 2020; 20: 533–34

Figure 6: Research work 4

The article by (Ensheng Dong, 2020) presents the concept of a global COVID-19 dashboard and details how the country-specific and global data on daily cases are gathered and formatted. This paper is closely connected to my project since the data sets that I got (worldwide-aggregate.csv and countries-aggregated.csv) are organized in a most similar manner. The study establishes that these datasets are dependable and they are common in scholarly investigations of the COVID-19 condition as well as prediction.

This confirms the quality of the data I will be using in my project, and it can be used to supervised machine-learning modelling.

Analysis

The article written by Ensheng Dong et al. (2020) is beneficial to COVID-19 prediction and this project because it defines the reliability and standardized format of the world and country-level daily COVID-19 data utilized in forecasting the data. The usage of datasets to be used in this project, including the worldwide-aggregate.csv and countries-aggregated.csv, is directly justified by the research by explaining how case data are regularly gathered, checked, and structured. Such validation guarantees that the time-series data that are being used in supervised learning and regression models are reliable, and that the generated COVID-19 predictions methodology is sound and comparable to the previous academic studies.

Machine Learning for COVID-19 Forecasting (Survey Paper)

url: <https://ieeexplore.ieee.org/document/9175487>



The screenshot shows the IEEE Xplore website interface. At the top, there's a navigation bar with links to IEEE.org, IEEE Xplore, IEEE SA, IEEE Spectrum, and More Sites. A search bar is present with a dropdown menu set to 'All'. Below the navigation bar, the article title 'Active Endoscope Preserving Image Orientation for Endonasal Skull Base Surgery' is displayed. The publisher is listed as IEEE, and there are buttons for 'Cite This' and 'PDF'. The authors are Geunwoong Ryu, Chulmin Park, Jeongryul Kim, and Keri Kim. The article has 151 full text views. The abstract section is visible, starting with 'Endonasal Skull Base Surgery has some advantages over conventional methods; however, some issues such as the need to replace the endoscope due to the fixed view, and possible invasiveness during insertion make the procedure difficult. To solve these problems, this paper proposes an active endoscope mechanism with several benefits. First, its variable direction of view can cover the wide ranges. Second, it provides a forward view, lessening the danger of damaging critical tissues while the endoscope is being inserted. Third, it can change the direction of view within a very small radius of 6.5 mm, so it can be used in a small cavity and work together with other surgical tools. We have designed the endoscope, which has 4 mm diameter. It is a simple tilting and rotation mechanism of 2-degree of

Figure 7: Research work 5

The article by (Shinde, 2020) is a survey of machine learning models applied in COVID-19 forecasting. It validates that the most common methods of predicting the trends of pandemics are supervised learning and regression models. Through the evaluation of the advantages and disadvantages of different forecasting methods, the paper assisted me in selecting suitable models including Linear Regression and Random Forest Regression to use in my project. This study will help me in building the methodological basis of my study and demonstrate that the techniques employed in my research are consistent with academic findings.

Analysis

The Shinde survey assists in predicting COVID-19 and the current project because it proves that supervised learning and regression models are some of the most effective and frequently used methods of forecasting the pandemic. Through the comparison of the strengths and weaknesses of different machine learning methods, the study gives a clear explanation why different models like the Linear Regression and Random Forest Regression have been chosen in this study. This directly proves the methodological decisions made in the project and indicates that the techniques used to predict the future are scientifically proven and comply with the traditions of scientific research in predicting the pandemic trends of COVID-19.

2.2 Review and Analysis of Existing Work

Existing Research Studies Using Similar Data

Data in this project has been used to predict and analyze COVID-19 trends, as well as by researchers in other studies. The key points of interest are trend analysis, forecasting, country-level comparisons and machine-learning-based prediction.

a) Trend Analysis and Forecasting.

Our everyday data, like those in `global-aggregate.csv`, are typically used to monitor the increase of the cases, observe outbreak-waves, and forecast possible future developments. It was demonstrated in early stages that COVID-19 case development was exponential or logistic, based on the pandemic stage (Wu, 2020). These findings uphold the application of the time-series analysis to the study of pandemic behavior.

b) Comparison Studies on a Country Level.

Such datasets as `countries-aggregated.csv` can be used to compare the level of infection, the fatality rate, and the effects of the government measures in different countries. The results of similar data studies have determined that the disparity among nations is mostly predetermined by the capacity of healthcare, testing policy, and the time of public-health interventions (Hale, 2021).

c) Prediction Studies of machine learning.

The features of machine-learning studies include the number of confirmed cases, recoveries, and deaths each day to forecast the numbers of the next day, recovery patterns, and mortality. Growth rate measures, like the increase rate feature of this data, are common to relate the short-run shifts in transmission and enhance model sensitivity to fast changes (Tuli, 2020).

🚦 **Current Projects, and Real-life implementations of similar data.**

🚦 **WHO Daily Situation Reports**

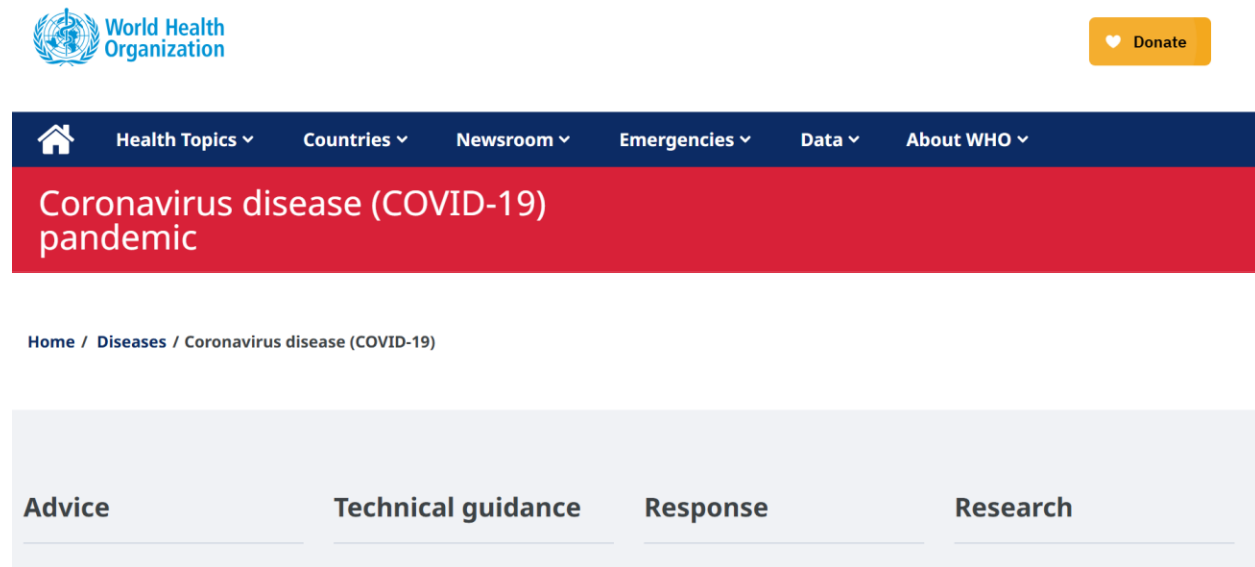


Figure 8: Existing work 1

(WHO, 2020) publishes daily situation reports which summarize the information on COVID-19 both globally and by country. The reports have confirmed cases, deaths, and recoveries. They are now a reliable source of aggregated information to monitor the pandemic and make decisions on the health of the population. The data of these reports corresponds to the format of the world-aggregate.csv and countries-aggregated.csv files which I use in my project.

Analysis:

These WHO daily reports are equivalent to the datasets in my project (e.g., worldwide-aggregate.csv) since both have aggregated data on cases in the world and in each country. They provide real time tracking and have been instrumental in predicting and decision making in the pandemic. As my project is based on the same types of data (daily confirmed cases, deaths, and recoveries), it is constructed on reputable publicly available information that is actively used to model the pandemic in real-time. This relationship increases the validity and applicability of my predictions.

COVID-19 Data Repository: Johns Hopkins University

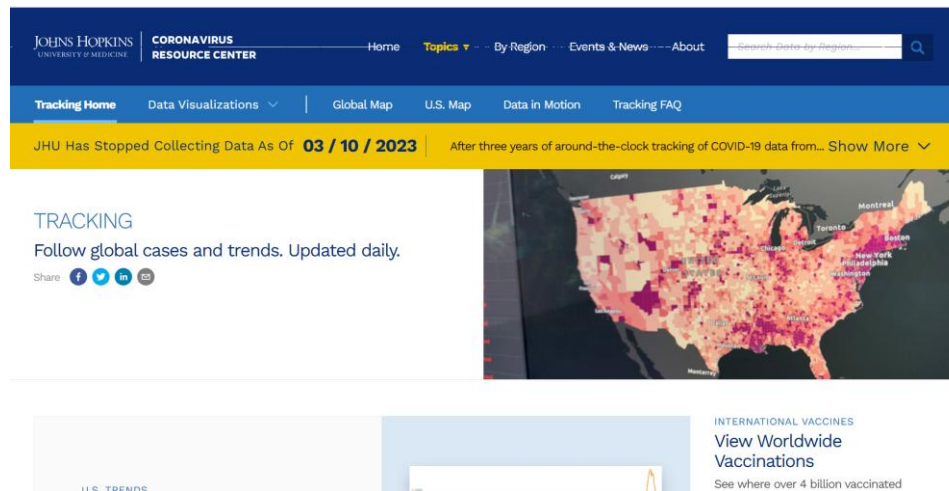


Figure 9:Existing work 2

Global and country-level cases were provided by this data set, which has been utilized in the development of COVID-19 forecasting models. It has been used by researchers worldwide to develop predictive models based on real-time case data (HopkinsUniversity, 2020).

Analysis:

JHU COVID-19 Data Repository provides the real-time data required in order to make precise forecasts. The format of the data is the one you use on your project hence a credible source of time-series forecast.

 Google COVID-19 Mobility Reports

🔍 google.com/covid19/mobility/



See how your community moved differently due to COVID-19

As global communities responded to COVID-19, we heard from public health officials that the same type of aggregated, anonymized insights we use in products such as Google Maps would be helpful as they made critical decisions to combat COVID-19.

These Community Mobility Reports aimed to provide insights into what changed in response to policies aimed at combating COVID-19. The reports charted movement trends over time by geography, across different categories of places such as retail and recreation, groceries and pharmacies, parks, transit stations, workplaces, and residential.

The Community Mobility Reports are no longer being updated as of 2022-10-15. All historical data will remain publicly available.

Community Mobility Reports

Reports created 2022-10-17.

In order to download or use the data or reports, you must agree to the Google [Terms of Service](#).

[Global CSV](#)

[Region CSVs](#)

[CSV documentation](#)

[How to use these reports](#)

Figure 10: Existing Work 3

The COVID-19 Mobility Reports of Google examine the location data of mobile devices. They monitor the dynamics of human movement due to pandemic-related restrictions and allow us to observe how mobility is related to case spikes and how interventions like lockdowns work (Google, 2020).

Analysis:

These mobility reports are supplementary to the case data utilized in this project, as they provide more background information about public health interventions. Mobility data would also be valuable to augment the accuracy of forecasting because it considers the association between human behavior and virus transmission that could be used to improve pandemic forecasts.

Time series forecasting of COVID-19 transmission in Canada using LSTM networks



Chaos, Solitons & Fractals

Volume 135, June 2020, 109864



Time series forecasting of COVID-19 transmission in Canada using LSTM networks ☆

Vinay Kumar Reddy Chimmula , Lei Zhang

[Show more](#) ✓

[+](#) Add to Mendeley [Share](#) [Cite](#)

<https://doi.org/10.1016/j.chaos.2020.109864> ↗

[Get rights and content](#) ↗

● Complimentary access

Highlights

- A fully automated, real-time forecasting model for COVID-19 transmission to help frontline health workers and government policy makers.
- Use of Artificial intelligence (AI) and Deep Learning to model Infectious diseases without losing temporal components.

Figure 11: Existing work 4

In the study, Long Short-Term Memory (LSTM) models are employed to predict COVID-19 cases in real-time, which is a deep-learning forecasting method of time-series. It compares LSTM with the traditional approaches like ARIMA (Chimmula, 2020).

Analysis:

The article establishes that time-series forecasting models such as LSTM are capable of predicting the spread of COVID-19. Though your project relies on regression models, the demonstration of the comparison with deep-learning methods helps to support the legitimacy of your approach.

Critical Analysis of Existing Work (Based on Gaps in my Data)

Although these datasets are useful, several limitations exist, and most of them have been pointed out in earlier studies. On the one hand, demographic data (age, gender, comorbidities) are not provided, which restricts the possibility of analysing the severity and vulnerability of the population. Second, the absence of the policy and intervention data, including lockdowns, mask requirements, travel bans, and vaccination rates, complicates the explanation of the reasons behind the growth or decline of the number of cases. Third, incorrect reporting of irregularities, such as delayed updates and corrections of the data, introduce noise in the data set, and machine-learning models should address it cautiously. Moreover, no data about the variants of the virus or even the implementation of vaccination are presented in the dataset, which became crucial in the further scenario of the pandemic. Lastly, the high-resolution analysis with hourly or region-level data is constrained by the use of daily aggregated data.

3. Solution

3.1 Proposed Solution / used AI algorithm

This project aims to create a machine learning forecasting system capable of predicting the number of cases of COVID-19 in the world with the help of supervised regression. It uses historical data in the form of worldwide-aggregate.csv and countries-aggregated.csv to feed the model to transform the raw case counts into insights that demonstrate how the virus spread during various stages of the pandemic. The system also derives timeseries characteristics, such as lag values, rolling averages and growth rates, to assist the models in capturing both the long-term trends and short-term variations. By using this method, the forecasting engine can capture the real-world COVID-19 behaviors, which frequently varies as a result of new variations, public health initiatives, and reporting modifications.

The forecasting process is aimed at identifying trends in the number of cases confirmed, recovered, and died daily and then relying on the trends to predict the number of cases to come. The model is trained on historical data to understand the impact of the past values on the future trend, and thus it can be used to estimate whether the cases will increase, stabilize, or decrease soon. Such predictions allow researchers and health authorities to anticipate the risk periods and prepare in advance, including the admission of more patients, the initiation of vaccination campaigns, and the redistribution of resources during the high transmission.

To address the complexity of COVID-19 data, the project compares various supervised regression models such as Linear Regression, Random Forest Regression, Linear Regression provides a bare minimum of the trends, whereas the nonlinear relationships of the data and the ability to deal with noisy or irregular data are offered by Random Forest. Such metrics as MAE, RMSE, and MAPE are used to evaluate model performance, and the most effective algorithm is chosen to make reliable short-term

predictions that will help to make decisions based on data and plan the work of the entire population to improve the health of the population.

Algorithm Used

The suggested solution includes three machine learning models predicting trends of COVID-19 cases according to past epidemiological data and designed time-series characteristics. The selected models are:

❖ **Linear Regression**

Linear Regression is a supervised algorithm in learning, and it is commonly applied in predicting continuous numeric values. It is especially appropriate in forecasting cases of COVID-19 since it is the model of the relationship between independent variables (including the past numbers of cases, growth rates, rolling averages) and the dependent variable (including the number of confirmed cases of COVID-19 in the future). The algorithm is applied by fitting a linear equation that reduces the prediction error of actual versus predicted values (Montgomery, 2012).

Mathematical Expression:

Linear Regression is a model that approximates a continuous dependent variable with linear equation of features that are continuous:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon$$

Where:

- y = predicted future COVID-19 cases
- $x_1, x_2, \dots, x_n$ = features (lag values, rolling averages, growth rates)
- β_0 = intercept
- $\beta_1, \beta_2, \dots, \beta_n$ = coefficients (impact of each feature)
- ε = error term

Figure 12: Numerical of Linear

Estimation of the model parameters is done through the minimization of the Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Figure 13: Mean Squared Error

Conceptual Figure

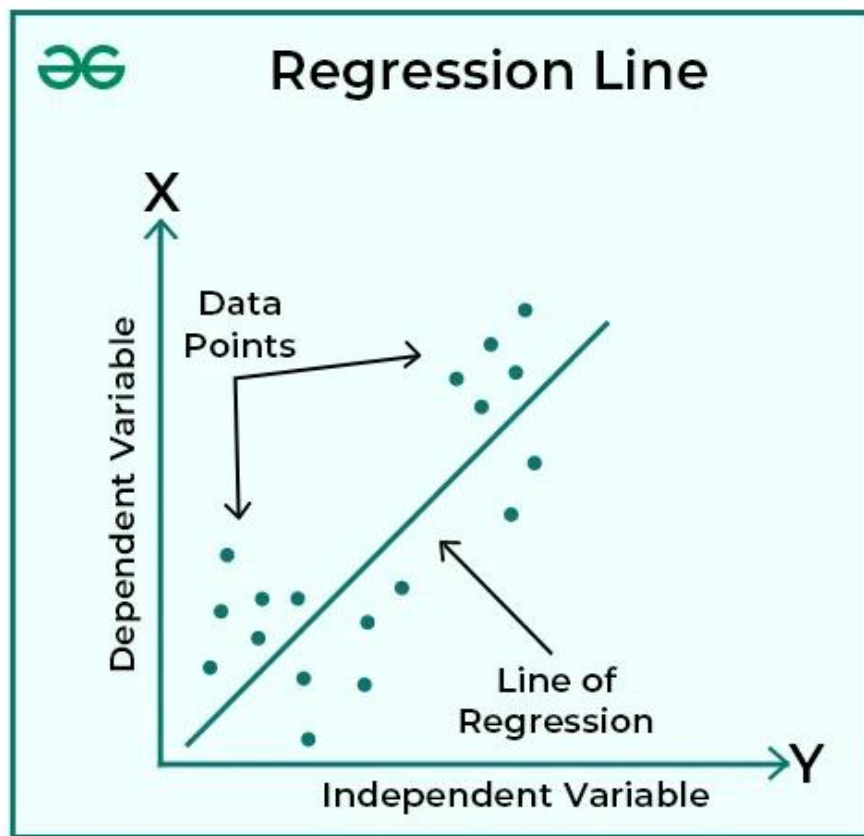


Figure 14: Conceptual figure 1

Relevance to COVID-19

Linear Regression is a method to indicate the general case growth tendency and can be used as a baseline to determine improvements by more advanced models.

Linear Regression is chosen in the given project due to its simplicity, chance to understand, and potency to determine general tendencies in the pandemic. In this model, a number of properties render it suitable in COVID-19 prediction, which are:

- It offers a distinct starting point when deciding how the overall COVID-19 cases go up and down.
- It is computationally cheap, and it can be trained and predicted on large datasets in a short time.
- It is very interpretable giving health analysts insight into the effects of each feature on predictions.
- It is good when variables are more or less linear in relationships.

These benefits make Linear (Hyndman, 2018)Regression an underlying model to describe the overall development of the COVID-19 cases and compare the results with more sophisticated models.

❖ Random Forest Regression

Random Forest Regression is an ensemble learning algorithm that is supervised and builds several decision trees, and the outputs of these trees are integrated to give a final output. This model is especially useful when forecasting COVID-19 since pandemic data in the real world is usually noisy, with sudden spikes and nonlinear relationships that are not well represented by simple linear model forecasts (Breiman, 2001).

The strengths of the Random Forest model that makes it the preferred choice in this project are the following:

- It takes the forecast of several decision trees thus being more accurate and stable.
- It also helps to minimize overfitting characteristics of the single decision-tree models through averaging of the results over the trees.
- It is able to process nonlinear trends and complicated interactions that exist in COVID-19 time-series data.
- It works well with high dimensional features (lag values, rolling averages, growth rates etc).
- It gives an importance score of features and thus, it can be used to determine the factors that significantly contribute to the prediction of the cases being considered.

Random Forest Regression is a strong and effective model that can be used to predict trends of future COVID-19 cases due to its strength and capacity to deal with unpredictable patterns of reporting.

Mathematical Expression:

Random Forest Regression is a combination of several decision tree predictions:

Figure 15:Formula 1

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x)$$

Where:

- T = number of decision trees
- $f_t(x)$ = prediction from the t^{th} tree
- $\hat{y}$ = final averaged prediction

Figure 16: Formula Description 1

Conceptual Figure

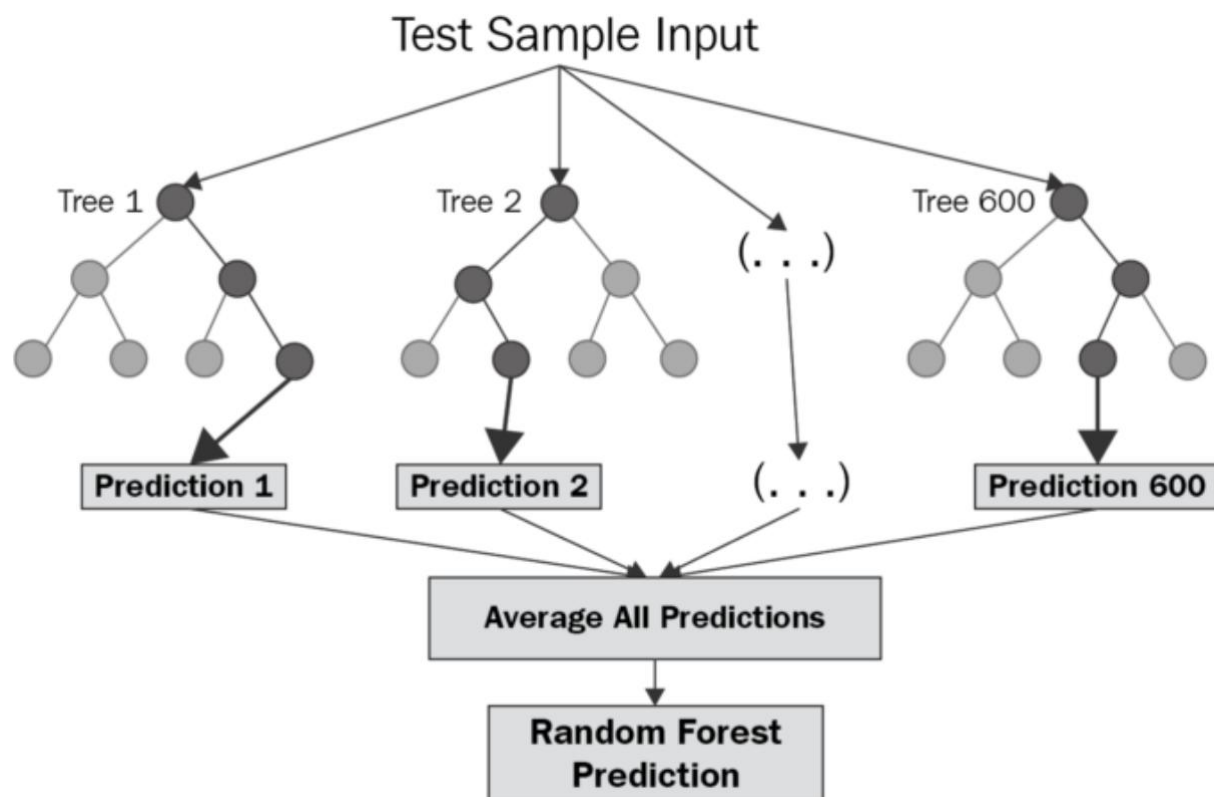


Figure 17: Conceptual Figure 2

Relevance to COVID-19

Random Forest can process non-linear trends, noise, and unforeseen spikes in COVID-19 reporting hence it is appropriate to real-life pandemic data.

❖ Feature Engineering Time-Series Regression.

A time-series regression is a supervised learning method, which considers the time aspect of a sequence of data. As is highly likely, COVID-19 case data should be seen as a time series simply because it is measured on a daily basis. The time-series regression methodology is applied in the present project by adding engineered time-based features into the standard regression models (Hyndman, 2018).

Mathematical Expressions

Engineered features are provided instead of modeling the raw time only:

Lag Features

Previous day confirmed cases (These cases are not counted in the daily tally of reported cases)

$$Lag_k = y_{t-k}$$

Figure 18:Lag Features

Rolling Average

Smooths daily fluctuations

$$RA_n = \frac{1}{n} \sum_{i=0}^{n-1} y_{t-i}$$

Figure 19: Rolling Average

Growth Rate

Shows acceleration or decline in spread

$$GR_t = \frac{y_t - y_{t-1}}{y_{t-1}}$$

Figure 20:Growth Rate

The following features are then taken as inputs to the regression models:

$$y_{t+1} = f(Lag_1, Lag_2, RA_n, GR_t)$$

Figure 21: Time-Series Feature-Based Regression formula

Conceptual Figure

Date	Temp		Temp	Lag 1	Lag 2	Lag 3
1981-01-01	20.7		20.7	NaN	NaN	NaN
1981-01-02	17.9		17.9	20.7	NaN	NaN
1981-01-03	18.8		18.8	17.9	20.7	NaN
1981-01-04	14.6		14.6	18.8	17.9	20.7
1981-01-05	15.8		15.8	14.6	18.8	17.9

Figure 22: Conceptual Figure 3

Relevance to COVID-19

Time-series regression also allows the models to acquire the temporal dependencies, enhancing the short-term forecasting accuracy when used in combination with Linear and Random Forest Regression.

The strategy works because of the following factors:

- Lag features (past day case counts) assist the model to comprehend short term relationships.
- Rolling averages even out reporting anomalies, and smooth out the noise on day-to-day data.
- Growth rate characteristics show an increase or a decrease in the spread of the virus.
- It enhances the accuracy of forecasting because it incorporates past trends as model inputs.

Time-series regression enables the models to be trained on how the past trends of the infection affect the future and thus it is very applicable in the forecasting of pandemics. This method has been combined with Linear Regression and Random Forest Regression to improve predictive performance and reliability.

3.2. Pseudocode

START

Load COVID-19 datasets (worldwide-aggregate.csv, countries-aggregate.csv)

Perform data cleaning

- Remove missing values

- Convert date fields to time-series format

Generate time-series features

- Calculate lag values (previous day cases)

- Compute rolling averages (3-day, 7-day)

- Calculate percentage growth rate

Split dataset into training and testing sets

Initialize machine learning models

- Linear Regression model

- Random Forest Regression model

Train models using training data

Evaluate models using test data

- Calculate MAE

- Calculate RMSE

- Calculate MAPE

Compare evaluation results

Select the model with better performance

Accept user input for recent COVID-19 case data

Prepare input features using the same feature engineering steps

Predict next-day confirmed cases using trained models

Display predicted results

END

3.3. Diagrammatical representations of the solution

Here are the diagrammatical representations of the project:

➤ **Flowchart**

This figure gives a step-by-step flowchart of the proposed AI-based COVID-19 prediction system. The workflow begins with the loading of the global COVID-19 datasets and then a full data cleaning process is implemented to handle missing data, date format and outliers. The feature engineering is done to fit time trends with lag features, rolling averages, and growth rates. The data is then divided into the training and the testing sets and two supervised machine learning models (Linear Regression and the Random Forest Regression) are trained simultaneously. The evaluation of the models is performed based on the traditional performance metrics, i.e. MAE, RMSE, MAPE. A decision block is used to compare the model errors of the two models in order to identify the most performing one. Using the selected model, final forecasts are then created by accepting recent user inputs and then preparing features in the same engineering process. In this flowchart, it is possible to see how the entire pipeline of the end-to-end is presented, and the model comparison strategy applied in the project.

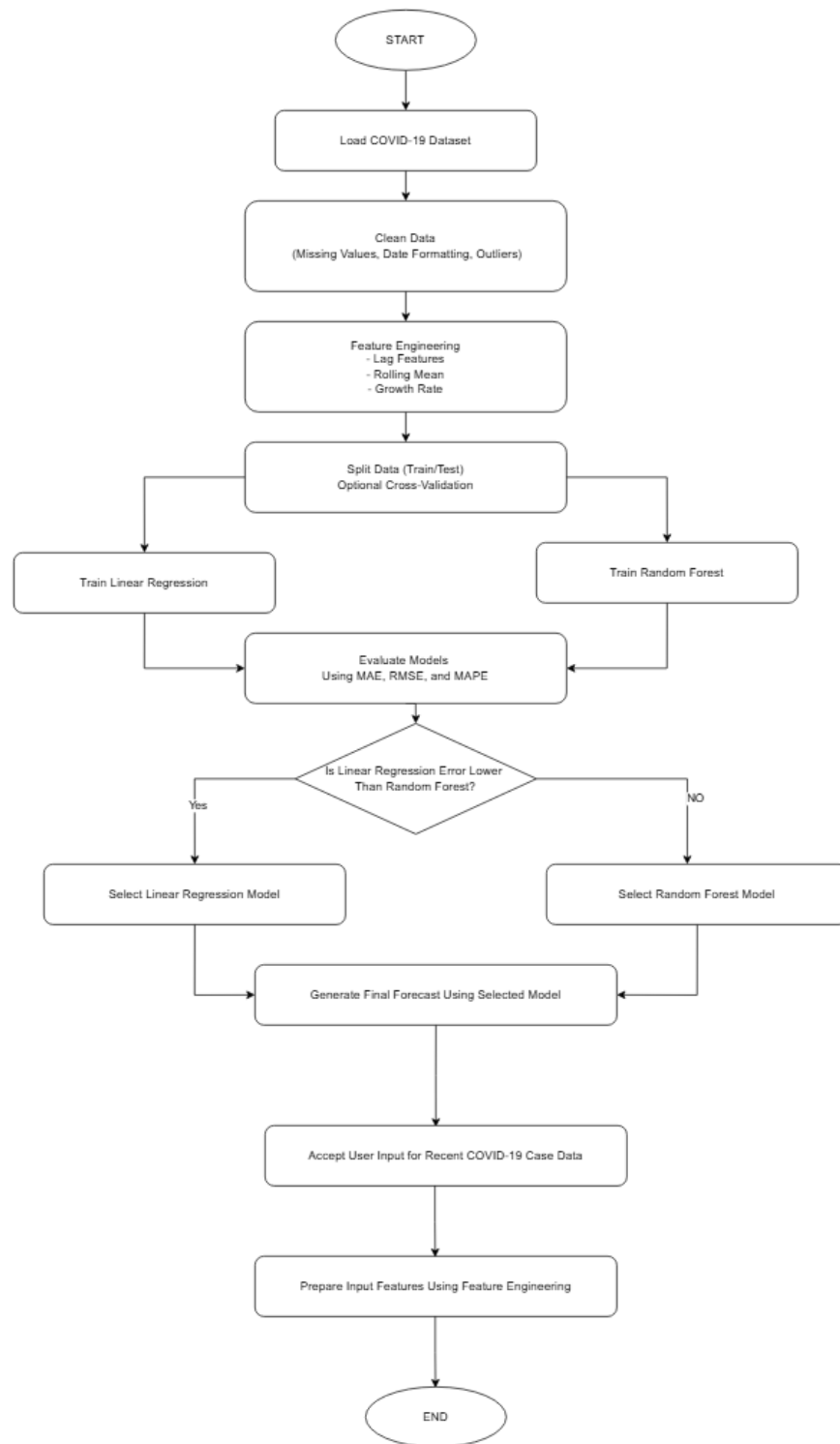


Figure 23:Flowchart of the project

➤ State transition diagrams



Figure 24: State transition diagrams

This figure represents a simplified workflow of the COVID-19 prediction system that has been created during this project. The data is loaded with the COVID-19 dataset and then undergoes the data preprocessing to clean up the data including missing values and preparing the data in time-series format. The feature engineering is then conducted to obtain meaningful time-dependent features such as lag values, rolling mean, and growth rate to assist the model in how the number of cases used to be in the past. The supervised machine learning models are developed and tested with the standard performance measures like MAE, RMSE, and MAPE after feature extraction. According to these assessment findings, the model having the minimum prediction error is chosen. The chosen model then takes updated COVID-19 case data provided by the user, uses the identical feature engineering procedure, and creates a forecast of the upcoming case of COVID-19 cases the following day. On the whole, the diagram presents a high-level view of the system workflow and demonstrates the flow of data (between the preprocessing phase and final prediction) in a easily understandable way.

3.4. Development process and Tools

To implement the COVID-19 prediction system successfully, there was a combination of the existing data science tools and libraries in order to ensure the good work in terms of the work with dataset management, the train of the machine learning model, the evaluation and the results visualization. These tools helped in general development cycle which is the data preprocessing to the predictive analysis.

❖ Anaconda



Figure 25: Anaconda

Anaconda is a high-level Python distribution, which is designed to be applied to data science and machine learning applications. It simplifies the process of package management and dependency management as it provides a pre-configured environment to make sure that all the libraries needed are compatible. In this project, all the required Python packages were installed and controlled with the assistance of Anaconda. Pandas, NumPy, Scikit-Learn, Matplotlib and Seaborn were installed in this project. Anaconda was employed to ensure a stable and isolated environment of the development process to reduce configuration issues and improve the reproducibility of the results (Anaconda, 2024).

❖ Jupyter Notebook



Figure 26:Jupyter notebook

This project was primarily developed using Jupyter Notebook which is included in the Anaconda distribution. It has permitted interactive execution of Python code, which has enabled an incremental testing of data preprocessing and feature engineering actions, and machine learning models. The use of code cells and Markdown documentation enabled the explanation of all the steps of the prediction process on COVID-19 in a clear way. In addition, Jupyter Notebook was able to provide real-time visualizations of the model outputs and evaluation metrics, this is why it can be employed in an exploratory analysis and interpretation of the results (Kluyver, 2016).

❖ Data visualization techniques.

- **Line Graphs**

Line graphs were used to visualize the changes of the COVID-19 confirmed cases and recoveries, as well as the deaths over a time. Line plots helped in that the data is a time-series type of data with a record being taken daily, therefore it was important to establish the waves of the infection, the peaks, and the downward trend thereafter. These graphical indicators were employed to support the decision making in feature engineering such as adding lag variables and rolling averages (Hyndman, 2018).

❖ Heat Maps

Correlations of numerical characteristics were also presented using heat maps i.e. confirmed cases, deaths, recoveries and growth rates. With a visual representation of the strength of correlation, the heat maps allowed seeing that there were strong dependencies between variables and the degree of multicollinearity. The analysis assisted in selecting important features to be used in prediction models by regression (Wilkinson, 2009).

❖ Python Libraries Used

❖ Pandas



Figure 27:Pandas

Pandas was popular in data loading, cleaning and preprocessing. Pandas Data Frames were efficient in their operations such as working with missing values, removing duplicates, calculating rolling averages and building lag features. It was suitable since it had flexible data manipulation capabilities and hence could manage the massive COVID-19 data (McKinney, 2010).

❖ **NumPy***Figure 28: Numpy*

NumPy helped in the calculations involving numbers and array operations that were required during feature engineering and in developing the model. It enabled the capability of conducting mathematical operations on large numeric data which is paramount to the machine learning processes that operate on the principle of utilizing vectors (Harris, 2020).

❖ **Scikit-Learn (sklearn)***Figure 29: Scikit-Learn*

Scikit-Learn, which is a large library, was used in the development of the machine learning used in this project. It also provided the intensive applications of supervised regression algorithms such as the Linear Regression and the Random

Forest Regression. The library was also used to split datasets into training and testing sets, model evaluation and compare performance measures. The Skit-Learn was used to aid the predictive modeling task and was regularly simple to utilize because of its API and frequently high stability (Pedregosa, 2011).

❖ **Train-Test Split**

The dataset was divided using the training test split which consisted of 80 and 20 training and testing subsets respectively. This ensured that models are trained on the historic data and tested on unseen data to the objective of finding out the accuracy of prediction and generalization (Trevor Hastie, 2009).

❖ **Evaluation Metrics**

To measure the performance of the models, the following measures of evaluation were used:

Mean Absolute Error (MAE): This value is used to estimate the average absolute error of the forecast.

Root Mean Square error (RMSE): Pales predictions scandals of larger magnitude to a greater extent.

Mean Absolute Percentage Error (MAPE): Is an error measurement expressed as a percentage, and the findings can be understood in the real world.

Through these measures, a comparative analysis of the Linear Regression and the random forest regression model was conducted at a large scale (Chai, 2014).

❖ Matplotlib and Seaborn



Figure 30: Matplotlib and Seaborn

Matplotlib was used to generate the line plots and comparison charts of the actual and the predicted cases of COVID-19. More complicated visualisations (heat maps and distribution plots) were made with Seaborn, built on top of Matplotlib. The combination of these libraries increased the interpretation ability of findings via visualisation of trends, correlation, and model performance (Waskom, 2021).

❖ Language



Figure 31: Python

Python was the main programming language, as it is readable, flexible, and rich in libraries that helped in the implementation of data analysis and machine learning tasks. Data manipulation and numerical calculations were done using libraries like Pandas and NumPy and the Scikit-Learn was used to create and test machine learning models. Matplotlib and Seaborn were used to perform data visualization so as to present insights and model results in a clear manner. Python offers a dependable and effective platform to create, test and assess machine learning solutions (McKinney, 2010).

3.5. Development and Achieved results

The following section includes the experimental findings of the proposed COVID-19 prediction system, such as data preprocessing, feature engineering, model training, and evaluation, and final forecasting results.

❖ Model analysis and prediction.

❖ Import Libraries.

```
|: from IPython.display import display
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import time

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

sns.set()
```

Figure 32: The Libraries that are to be imported to predict the COVID-19.

The code includes a complete python environment to analyze data and machine learn by importing some of the essential libraries that give a chance to work with and analyze data, visualize the results and construct predictive models. It collaborates with Pandas and NumPy to do data manipulations and calculate numbers, Matplotlib and Seaborn to draw readable and attractive plots and Time to measure the time taken. It offers the functionality of scikit-learn library to separate datasets into training and test sets, run the regression models, including Linear Regression and Random Forest Regressor and evaluate the model performance in terms of the standard metrics, including Mean Absolute Error, Mean Squared Error, and R2 score, and sns.set offers similar visualization styling throughout the analysis.

❖ Load dataset.

```
#Data Load
world_df = pd.read_csv(
    r"C:\Users\Dell\Desktop\23048838 Muna Pun AI\worldwide-aggregate.csv"
)

country_df = pd.read_csv(
    r"C:\Users\Dell\Desktop\23048838 Muna Pun AI\countries-aggregated.csv"
)

world_df["Date"] = pd.to_datetime(world_df["Date"])
country_df["Date"] = pd.to_datetime(country_df["Date"])

# SHOW OUTPUT (IMPORTANT)
display(world_df.head())
display(country_df.head())
```

Figure 33:Loaded Datasets

The code reads COVID-19 data sets in the form of CSV files in a local directory, which are then analysed using Pandas DataFrames, with world_df having worldwide aggregated data, and country_df having country-level aggregated data. Once the datasets have been loaded with the help of the command `pd.read_csv`, the Date column of the two DataFrames receives the transformation into a proper datetime format with the help of the command `pd.to_datetime` to allow proper time-series analysis. Lastly, to quickly ensure that the data is loaded properly and is in the expected form and format, the `display()` function can be used to show the first few rows of each dataset.

Output:

	Date	Confirmed	Recovered	Deaths	Increase rate
0	2020-01-22	557	30	17	NaN
1	2020-01-23	657	32	18	17.953321
2	2020-01-24	944	39	26	43.683409
3	2020-01-25	1437	42	42	52.224576
4	2020-01-26	2120	56	56	47.529576

	Date	Country	Confirmed	Recovered	Deaths
0	2020-01-22	Afghanistan	0	0	0
1	2020-01-23	Afghanistan	0	0	0
2	2020-01-24	Afghanistan	0	0	0
3	2020-01-25	Afghanistan	0	0	0
4	2020-01-26	Afghanistan	0	0	0

Figure 34:Output of load datasets

The figure shows the first few rows of two COVID-19 datasets with the loading and preprocessing made. The first table is a global aggregated one, where the total amount of confirmed cases, recovered cases, death, and the rate of increase are indicated by date, where the initial date (2020-01-22) has no increase rate because there is no previously recorded value. The second table provides country-specific statistics of Afghanistan, the data of which include daily confirmed cases, recoveries and deaths, all of which are zero in the first dates provided, indicating that in the early period of the pandemic, there were no reported cases. The combination of the figures proves that the datasets are properly organized and ready to be analyzed as the global and country-wise time-series.

❖ Data overview.

```
: #Dataset Overview & Preview
overview_df = pd.DataFrame({
    "Dataset": ["Worldwide Aggregate", "Countries Aggregated"],
    "Rows": [world_df.shape[0], country_df.shape[0]],
    "Columns": [world_df.shape[1], country_df.shape[1]]
})
display(overview_df)

display(world_df.head())
display(country_df.head())
```

Figure 35:Data overview

The figure displays a Python code block that was utilized to give an overview of the COVID-19 datasets and preview it. It generates a summary DataFrame called `overview_df` which somehow compares the aggregate worldwide dataset and the countries aggregated dataset by showing their names and the total number of rows and columns they contain. This summary assists in getting the dimensions and framework of the datasets prior to examination. The first several rows of both `world_df` and `country_df` are then shown in the code, and one can quickly visually register the content of the data and ensure that the datasets have been loaded and formatted properly.

Dataset Overview Output:

	Dataset	Rows	Columns
0	Worldwide Aggregate	816	5
1	Countries Aggregated	161568	5

	Date	Confirmed	Recovered	Deaths	Increase rate
0	2020-01-22	557	30	17	NaN
1	2020-01-23	657	32	18	17.953321
2	2020-01-24	944	39	26	43.683409
3	2020-01-25	1437	42	42	52.224576
4	2020-01-26	2120	56	56	47.529576

	Date	Country	Confirmed	Recovered	Deaths
0	2020-01-22	Afghanistan	0	0	0
1	2020-01-23	Afghanistan	0	0	0
2	2020-01-24	Afghanistan	0	0	0
3	2020-01-25	Afghanistan	0	0	0
4	2020-01-26	Afghanistan	0	0	0

Figure 36:Dataset Overview Output

The figure provides an overview and preview of the COVID-19 datasets utilized in the analysis. The difference in the level of data granularity is emphasized in a dataset overview table presented at the top where the worldwide aggregate dataset (with 816 rows and 5 columns) is compared to the countries aggregated dataset (with 161,568 rows and 5 columns). The first few rows of the global dataset are presented below and consist of date-specific global confirmed cases, recoveries, deaths, and the calculated increase rate, the oldest date has an empty increase rate the reason being that there is no previous data. The last table presents the preview of the country-level data on Afghanistan, in which confirmed, recovered, and deaths are all zero in the first few dates, which indicates the initial phase of the pandemic.

❖ Aggregate Country Data

```
#Aggregate & Merge Datasets
country_agg = country_df.groupby("Date").agg({
    "Confirmed": "sum",
    "Deaths": "sum",
    "Recovered": "sum"
}).reset_index()

country_agg.columns = [
    "Date", "Country_Confirmed", "Country_Deaths", "Country_Recovered"
]

merged_df = pd.merge(world_df, country_agg, on="Date", how="inner")
merged_df = merged_df.sort_values("Date").reset_index(drop=True)

display(merged_df.head())
```

Figure 37: Aggregate Country Data

The figure presents a data processing phase, in which country-level data on COVID-19 is combined and integrated with global data to compare them. The dataset by country is then group by date and the number of confirmed, death, and recovery cases in all countries is summed up. These columns are then renamed to ease the understanding and then the columns are joined together with the global aggregate data, on the shared date field, through inner join. Lastly, the merged data is ordered in time-order and previewed, so that both global and aggregated country-level statistics can be identical in date, and again usable in additional analysis or modeling.

Output:

	Date	Confirmed	Recovered	Deaths	Increase rate	Country_Confirmed	Country_Deaths	Country_Recovered
0	2020-01-22	557	30	17	NaN	557	17	30
1	2020-01-23	657	32	18	17.953321	657	18	32
2	2020-01-24	944	39	26	43.683409	944	26	39
3	2020-01-25	1437	42	42	52.224576	1437	42	42
4	2020-01-26	2120	56	56	47.529576	2120	56	56

Figure 38:Aggregate data output

The figure illustrates a preview of the merged COVID-19 data that is a combination of global statistics plus aggregated country level data on a date-by-date basis. Every row

will be a certain date and will contain the global confirmed cases, recoveries, deaths, and the calculated increase rate as well as the respective totals obtained by adding all the countries data to the same date. The similarity between the values of the worldwide and country-aggregated values indicates an effective process of aggregation and merging whereas the lack of an increase rate in the first date shows the lack of the previous data to compare. Such consolidated data will give a coherent understanding of world trends of COVID-19 and will be utilized in additional studies and predictive modeling.

❖ Missing Value

```
#Missing Values Check
missing_df = merged_df.isnull().sum().reset_index()
missing_df.columns = ["Column", "Missing Values"]
display(missing_df)
```

Figure 39: Missing value

The figure provides a piece of code to test the absence of values in the unified COVID-19 dataset. It sums the number of missing values in each column by using isnull function and sum function and then reset the index to rearrange the results in a clear tabular format. The columns are renamed according to dataset attributes and the number of missing values respectively and the results are shown so that one can easily find out incomplete data that may need cleaning or preprocessing before further analysis or modelling.

Output:

	Column	Missing Values
0	Date	0
1	Confirmed	0
2	Recovered	0
3	Deaths	0
4	Increase rate	1
5	Country_Confirmed	0
6	Country_Deaths	0
7	Country_Recovered	0

Figure 40: Missing value output

The figure shows the findings of a missing values analysis of the combined COVID-19 data. It displays all the columns with the number of missing values in each column indicating that the tables contain no missing values except the increase rate column which

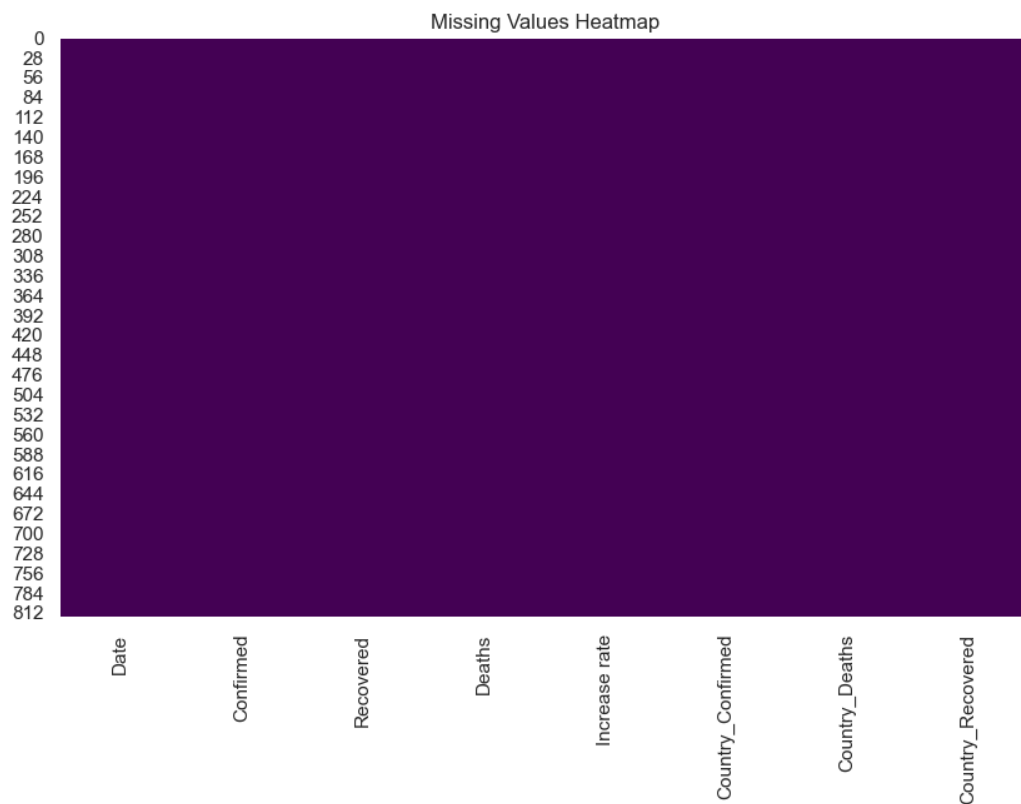
has a single missing value. This missing value is attributed to the first date in the dataset, on which the increase rate cannot be estimated because of lack of the previous data, which implies that the rest of the data are complete and can be used in future analysis.

❖ Missing Value Heatmap

```
#Missing Values Heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(merged_df.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Values Heatmap")
plt.show()
```

Figure 41:Missing value heatmap

The figure demonstrates a validation step where a new column, ConfirmedDiff, is created by subtracting the aggregated country-level confirmed cases (CountryConfirmed) from the worldwide confirmed cases (Confirmed) for each date. The output indicates that the difference between the two numbers is zero in all of the presented dates which means that the number of confirmed cases in the world is the same as the total number of confirmed cases in all the countries. This ensures that the process of aggregation and merge is correct and the two datasets are consistent, reliable and ready to be used in other analysis or predictive modeling.

Results:*Figure 42:Missing Values Heatmap***❖ Data Cleaning steps: handling missing value and removing duplicates**

```
#Handle Missing Values & Duplicates
merged_df.ffill(inplace=True)
merged_df.drop_duplicates(inplace=True)
merged_df.reset_index(drop=True, inplace=True)

display(pd.DataFrame({
    "Rows After Cleaning": [merged_df.shape[0]],
    "Columns After Cleaning": [merged_df.shape[1]]
})))
```

Figure 43:data cleaning steps

The figure shows the cleaning steps that were used on the combined COVID-19 data. The values that are missing are filled forward to ensure continuity of the time-series data and duplicates are eliminated, as well as the index restarts to make sure that data is correctly ordered in a sequence. This is done to make sure that the data is unified, complete, and can be analyzed or modeled further.

Results:

Rows After Cleaning		Columns After Cleaning
0		810
		14

Figure 44: Data size after cleaning

The figure gives the result of cleaning process in a summary table, which indicates that the dataset has 810 rows and 14 columns after cleaning. This shows that the duplicates that are not necessary have been eliminated and the structure of the datasets is completed giving a clean and trusted dataset to be used in future exploratory analysis and predictive modeling.

❖ Feature Engineering (Merged Dataset)

```
#Feature Engineering
merged_df["Lag_1"] = merged_df["Confirmed"].shift(1)
merged_df["Lag_2"] = merged_df["Confirmed"].shift(2)

merged_df["Rolling_3"] = merged_df["Confirmed"].rolling(3).mean()
merged_df["Rolling_7"] = merged_df["Confirmed"].rolling(7).mean()

merged_df["Growth_Rate"] = merged_df["Confirmed"].pct_change()

merged_df.dropna(inplace=True)
merged_df.reset_index(drop=True, inplace=True)

display(merged_df.head())
```

Figure 45: Feature Engineering

The figure shows the process of feature engineering that is applied to the combined COVID-19 dataset to make it more appropriate to time-series modeling. Lag variables (Lag1 and Lag2) are introduced to capture the temporal dependence in confirmed cases, rolling mean features (Rolling3 and Rolling7) are added to smooth the short-term and weekly trends; and a growth rate feature to measure the change in daily percent change in confirmed cases. Once these features are created, rows containing missing values

created by rolling and shifting operations are eliminated and the index is rearranged to ensure the correct alignment of data

Results:

	Date	Confirmed	Recovered	Deaths	Increase rate	Country_Confirmed	Country_Deaths	Country_Recovered	Lag_1	Lag_2	Rolling_3	Rolling_7	Growth_Rate
0	2020-01-28	5580	108	131	90.508706	5580	131	108	2929.0	2120.0	3543.000000	2032.000000	0.905087
1	2020-01-29	6169	127	133	10.555556	6169	133	127	5580.0	2929.0	4892.666667	2833.714286	0.105556
2	2020-01-30	8237	145	171	33.522451	8237	171	145	6169.0	5580.0	6662.000000	3916.571429	0.335225
3	2020-01-31	9927	225	213	20.517179	9927	213	225	8237.0	6169.0	8111.000000	5199.857143	0.205172
4	2020-02-01	12038	287	259	21.265236	12038	259	287	9927.0	8237.0	10067.333333	6714.285714	0.212652

Figure 46: Feature engineering output

The figure presents a preview of the dataset following feature engineering, which demonstrates the additional variables with the original COVID-19 attributes. It also shows how the lagged values, rolling averages, and growth rates are effectively combined with date-wise confirmed, recovered, and death counts to come up with a more comprehensive data that reflects historical trends as well as trend dynamics. This enriched data is very suitable to predictive modeling and regression analysis.

❖ Feature Engineering Histograms

```
#Feature Distribution
merged_df[
    ["Confirmed", "Lag_1", "Lag_2", "Rolling_3", "Rolling_7", "Growth_Rate"]
].hist(bins=25, figsize=(12, 8))
plt.suptitle("Feature Distribution Histograms")
plt.show()
```

Figure 47: Feature distribution

The figure demonstrates Python code that was applied to produce feature distribution histograms of the engineered variables of the COVID-19 data. It picks out important aspects like confirmed cases, lag values, rolling averages, growth rate, and displays their distributions in histograms of set bins. The same figure size and shared title are provided in order to provide uniformity and visual description of the distribution of all the features.

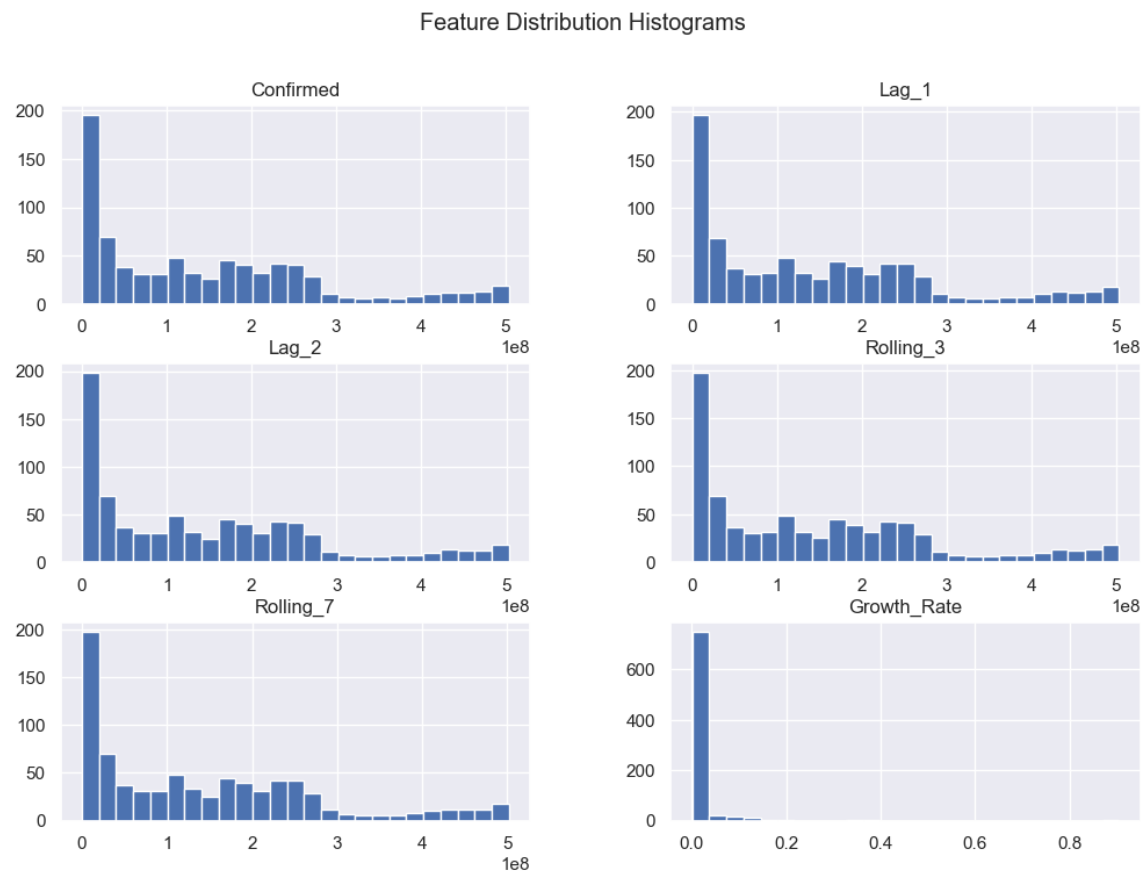
Results:

Figure 48: Feature distribution histograms

The figure will show the resultant feature distribution histograms, which show the distribution of each variable over the dataset. The distributions of the confirmed cases, lag features, and rolling averages are right skewed in nature, i.e. the small values are more common than the big values as the number of cases increases with time. The growth rate histogram is skewed towards zero indicating that the percentage changes in the daily percentage changes are not too large after the first outbreak stage. These visualizations combined aid in learning how the features behave and how they are prepared to do predictive modelling.

❖ Correlation Heatmap

```
#Correlation Heatmap
corr_features = [
    "Confirmed", "Lag_1", "Lag_2",
    "Rolling_3", "Rolling_7", "Growth_Rate",
    "Country_Confirmed", "Country_Deaths", "Country_Recovered"
]

corr_df = merged_df[corr_features].corr()
display(corr_df)

plt.figure(figsize=(12, 8))
sns.heatmap(corr_df, annot=True, fmt=".2f", cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Figure 49:Correlation heatmap

The figure shows the Python code used to compute and visualise the correlation structure between engineered and original COVID-19 variables. A smaller set of variables, such as the number of confirmed cases, lagged data, rolling averages, growth rates, and the aggregate country-level statistics are chosen carefully, and a correlation matrix is built on their basis. The script displays the tables of the correlation coefficients between the variables numerically and then makes a heatmap with the annotations of the numbers and colour gradient, thus explicitly showing the strength and direction of inter-variable relationships.

Results:

The resulting correlation table and heatmap demonstrate a high degree of positive correlation between confirmed cases, lagged values, rolling averages and country-level measures of confirmed and death cases, which suggest a high degree of temporal and structural dependency of the data. Conversely, the growth rate is showing a moderate negative correlation with these cumulative variables and it indicates that overall percentage changes are likely to decline with a number of total cases. The recovered case counts show poor relationships with the majority of the variables, which suggests another trend pattern. Taken together, such visualisations emphasize the multicollinearity between the time-dependent features and provide useful information in the selection of features and in the design of the model.

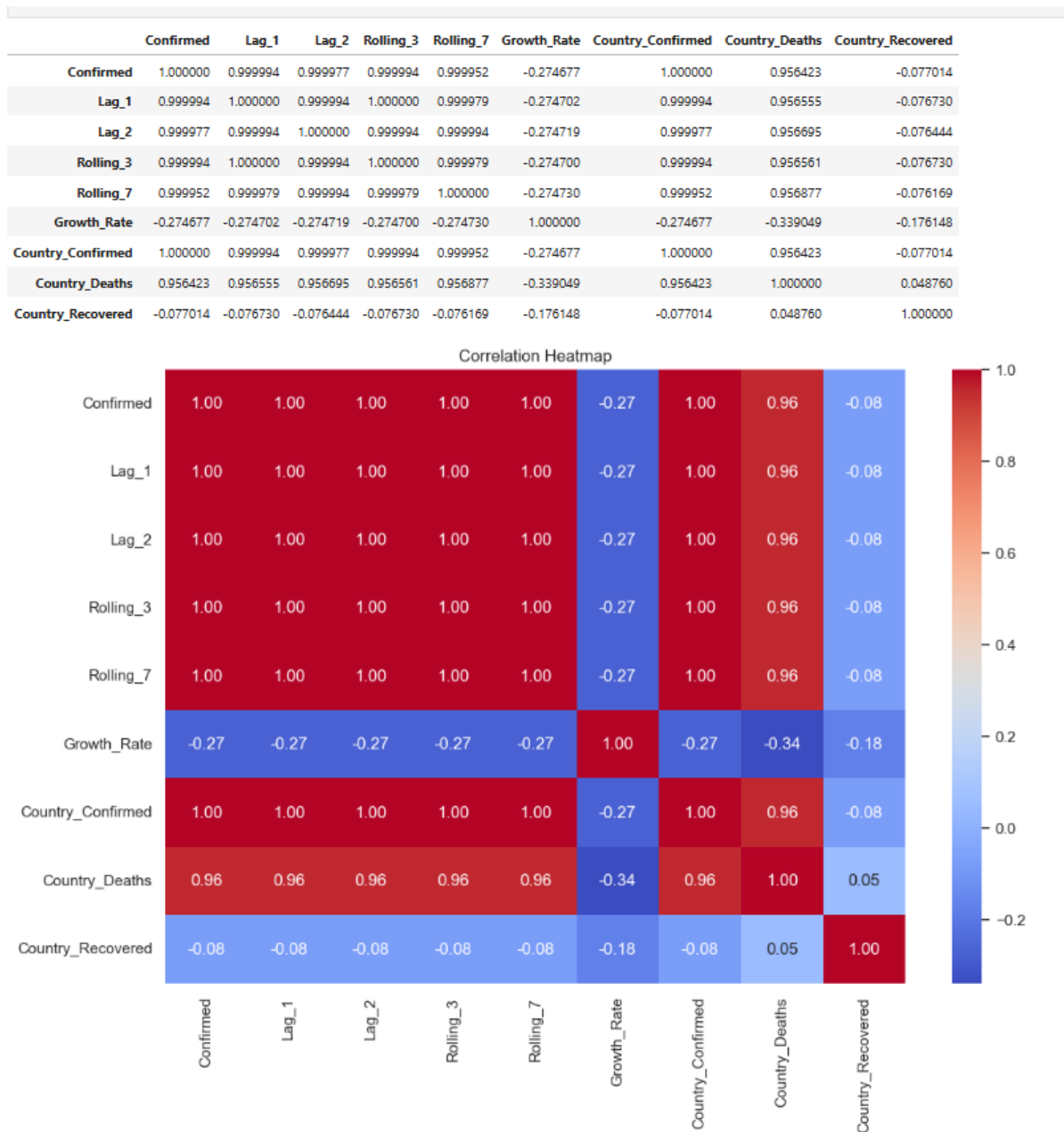


Figure 50: Result of correlation heatmap

❖ Feature Selection

```
: #Feature Selection & Train-Test Split
features = [
    "Lag_1", "Lag_2", "Rolling_3", "Rolling_7",
    "Growth_Rate", "Country_Confirmed",
    "Country_Deaths", "Country_Recovered"
]

X = merged_df[features]
y = merged_df["Confirmed"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, shuffle=False
)

display(pd.DataFrame({
    "Dataset": ["X_train", "X_test", "y_train", "y_test"],
    "Shape": [X_train.shape, X_test.shape, y_train.shape, y_test.shape]
})))
```

Figure 51: Selection of the features and train-test split

The figure X shows the process of feature selection and train-test division that was adopted in the COVID-19 prediction model. A set of relevant inputs variables such as lagged counts of confirmed cases, rolling averages, rates of growth, and aggregate country-level variables are assigned the explanatory variables and the cumulative count of confirmed cases is the response variable. Next, an 80-20 split of the data into training and testing subsets is performed without shuffling, thus maintaining the chronological order without which time series analysis would be impossible.

Results:

Below Figure is a summary table of the dimensions of the obtained training and testing sets. The table shows that the training set has 648 observations and eight predictor variables whereas the testing set has 162 observations and the same dimensionality of predictor variables; the corresponding response vectors have the same lengths. These outcomes affirm that the data are properly partitioned and are properly organized to train and test a model.

	Dataset	Shape
0	X_train	(648, 8)
1	X_test	(162, 8)
2	y_train	(648,)
3	y_test	(162,)

Figure 52: Result of feature selection

❖ Model Training and performance evaluation

```
# Model Training + Accuracy + Time
models = {
    "Linear Regression": LinearRegression(),
    "Random Forest Regression": RandomForestRegressor(
        n_estimators=100, random_state=42
    )
}

results = []
predictions = {}

for name, model in models.items():
    start = time.time()
    model.fit(X_train, y_train)
    train_time = time.time() - start

    y_pred = model.predict(X_test)
    predictions[name] = y_pred

    results.append({
        "Model": name,
        "MAE": mean_absolute_error(y_test, y_pred),
        "RMSE": np.sqrt(mean_squared_error(y_test, y_pred)),
        "MAPE (%)": np.mean(np.abs((y_test - y_pred) / y_test)) * 100,
        "R² Score": r2_score(y_test, y_pred),
        "Training Time (seconds)": train_time
    })

comparison_df = pd.DataFrame(results)
display(comparison_df)
```

Figure 53: Model training and performance evaluation

The figure shows the Python code used to train and test two regression models (Linear Regression and Random Forest Regression) on a COVID-19 dataset. Both models are trained on the training data, and the training time is recorded in order to be able to compare computational efficiency. After training, every model makes predictions on the test set and we calculate a number of evaluation metrics; MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), MAPE (Mean Absolute Percentage Error)

and R2 score. The indicators of the two models are recorded in a table of comparison, which is easy to interpret.

Results:

	Model	MAE	RMSE	MAPE (%)	R ² Score	Training Time (seconds)
0	Linear Regression	2.354751e-07	2.609056e-07	6.104283e-14	1.000000	0.015101
1	Random Forest Regression	1.152390e+08	1.453843e+08	2.734490e+01	-1.690206	1.053180

Figure 54:Result of model comparison results

These results confirm that the Linear Regression is the more efficient model to these properties of time-series and can be applied to the proper prediction of COVID-19 cases.

Evaluation Metrics Interpretation:

The measures of evaluation in this study, such as MAE, RMSE, MAPE and R2 score, are quantitative indications of accuracy of prediction of the models. The decreased MAE and RMSE values imply that the model is very accurate in terms of the closeness of the predicted values to the actual ones and a high R² value means that the model is very close to the actual values. These metrics play a crucial role in the context of COVID-19 prediction as the slightest error in prediction may result in the wrong interpretation of the severity of an outbreak and resource planning. Consequently, using a combination of measures will guarantee a stronger evaluation of model performance.

❖ Accuracy and Training Time Comparison

```
#Accuracy & Training Time Comparison
metrics = ["MAE", "RMSE", "MAPE (%)", "Training Time (seconds)"]

for metric in metrics:
    plt.figure(figsize=(8, 5))
    plt.bar(comparison_df["Model"], comparison_df[metric])
    plt.title(f"{metric} Comparison")
    plt.ylabel(metric)
    plt.show()
```

Figure 55:Accuracy and training time comparison

The figure displays a Python code segment comparing the accuracy and training time of various regression models through visual analysis. It will establish a list of evaluation measures, namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and training time, and will repeat the process with each of the measures to generate an individual bar chart. The models are placed on the x-axis, and the relevant values of metric are placed on the y-axis with the relevant titles and labels to make the models understandable. It is easy and intuitive to compare the model performance and computational efficiency under a variety of evaluation criteria in this visualization method.

Results:

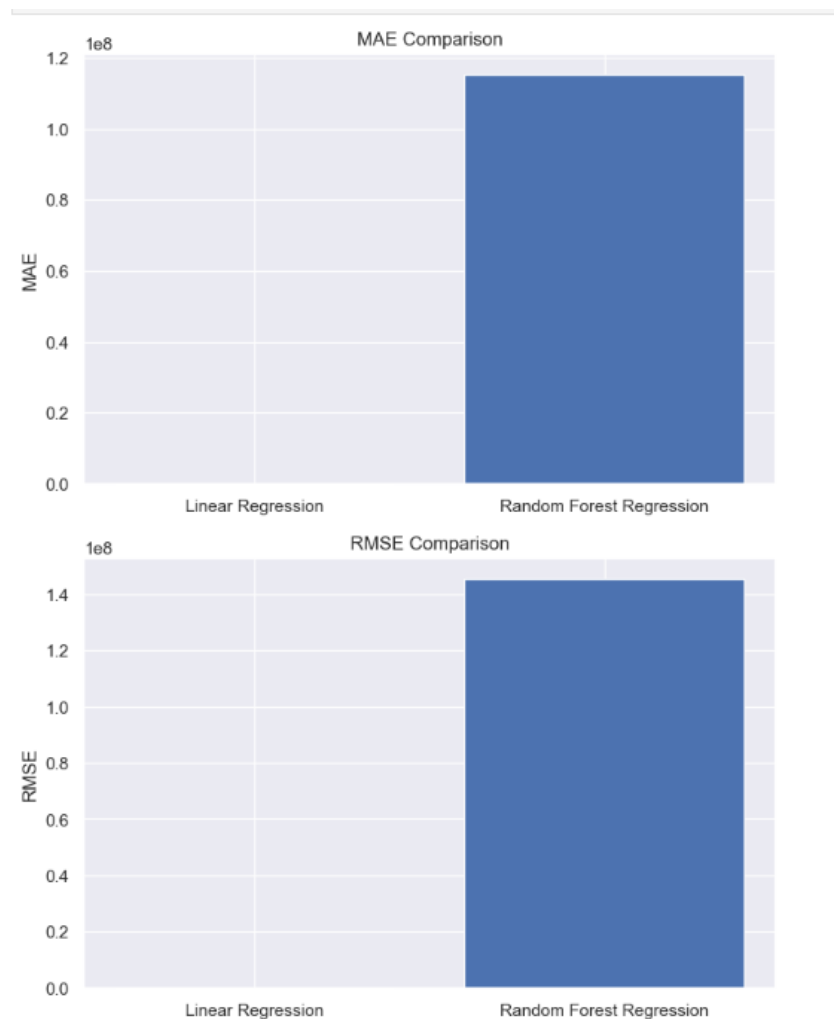


Figure 56: MAE and RMSE comparison

In the figure, there is a comparison of Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) of the Linear Regression and Random Forest Regression models. As it is evident in the bar charts, Linear Regression has nearly zero values of errors in MAE and RMSE, indicating that it is highly accurate in prediction. Conversely, the error values in the Random Forest model are significantly large indicating that it has worse predictive power on the COVID-19 time-series dataset.

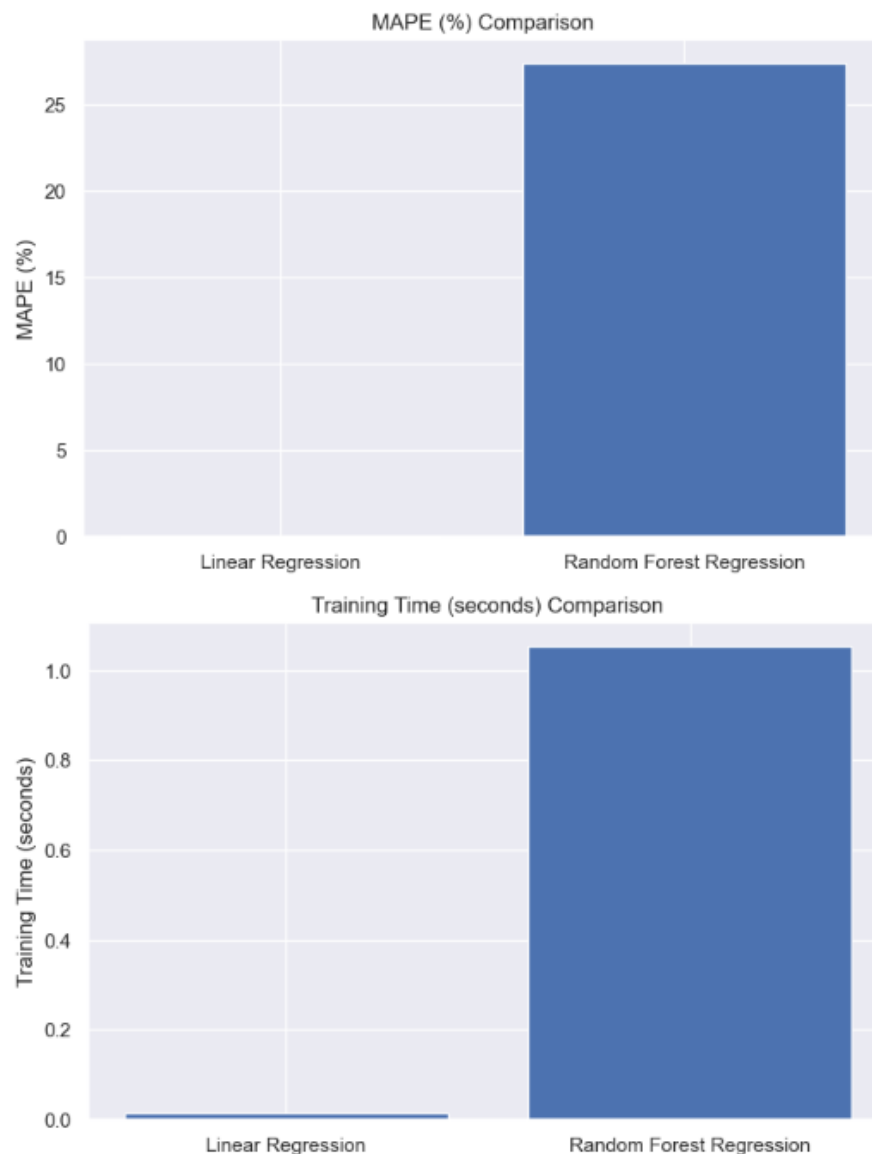


Figure 57: MAPE and training time comparison

The figure demonstrates the comparison of Mean Absolute Percentage Error (MAPE) and training time of both models. Linear Regression model is characterized by very small

MAPE, and it does not need much training, which underlines its accuracy and computational importance. Conversely, the MAPE of the Random Forest Regression model is far higher and the training time is considerably higher, which highlights the trade-off between model complexity and performance, which proves that Linear Regression is more fitting the current dataset.

This shows the compromise of model complexity versus model performance which demonstrates that Linear Regression is more appropriate to the present data.

Model Performance Differences

Discussion. In this study, Linear Regression was better than Random Forest Regression since the COVID-19 time-series data shows a very strong linear trend with time. Linear Regression can handle this structured and cumulative data reasonably well and, therefore, the nature of the underlying growth pattern can be captured. Random Forest Regression on the other hand is a non-linear ensemble technique which works better when the set of data is complex and highly varying, however, in this context, it could not generalize the temporal trend, and produced greater errors in predictions. This shows that the simpler models do sometimes work better than the complex models when the characteristics of the data would fit well with the linear assumptions.

❖ Actual vs Predict for model evaluation

```
#Actual vs Predicted
for name, y_pred in predictions.items():
    plt.figure(figsize=(12, 6))
    plt.plot(y_test.values, label="Actual")
    plt.plot(y_pred, label=f"Predicted ({name})")
    plt.title(f"Actual vs Predicted - {name}")
    plt.legend()
    plt.show()
```

Figure 58: Actual vs Prediction plotting code

The figure represents the Python code that plots the actual and predicted confirmed cases of COVID-19 using each of the trained regression models. The script reads through the predictions that have been stored and will plot the values of true tests and the values of

the prediction on the same graph. It then provides titles and legends such that one can easily compare the actual data with the results of the model. It is a visual aid that will allow you to determine the degree to which the predictions of each of these models fits the actual data over time.

Results:

The figure compares the real and forecasted tendencies of Linear Regression and Random Forest Regression. The regression line of Linear Regression plot intersects the actual curve near to the real curve meaning that it fits well and is accurate. In comparison, the Random Forest plot is much different; the line is nearly horizontal, and the real data increases, which shows that it is inefficient at reflecting the increase rate. This comparison in tabula rasa shows that Linear Regression is the most appropriate tool in this time-series forecasting task.

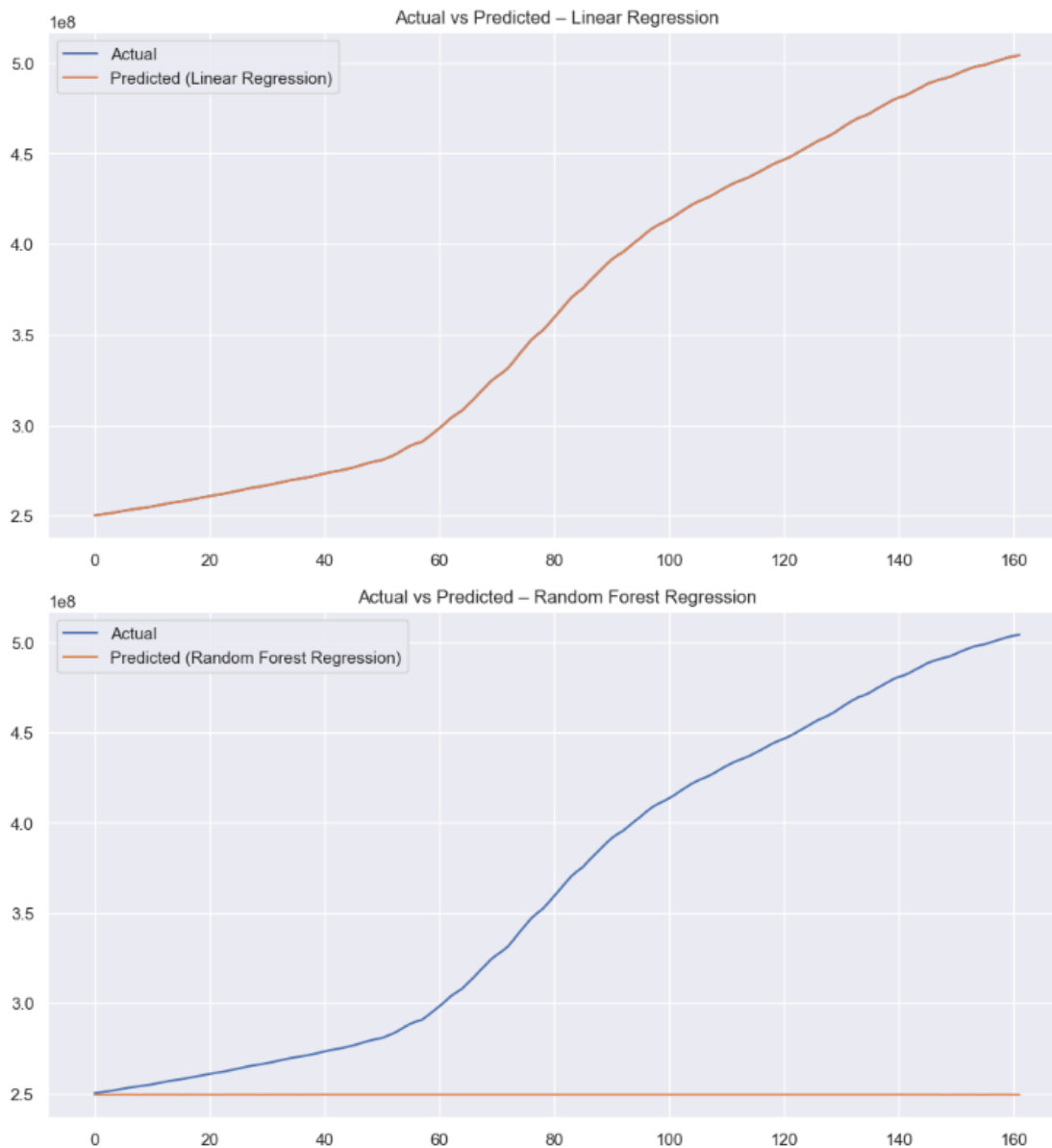


Figure 59: Actual vs Predicted COVID-19 Confirmed Cases: Linear Regression vs Random Forest

❖ Best Model Selection

```

#Best Model Selection
best_model_name = comparison_df.loc[
    comparison_df["RMSE"].idxmin(), "Model"
]
best_model = models[best_model_name]

display(pd.DataFrame({
    "Selection Rule": ["Lowest RMSE"],
    "Best Model": [best_model_name]
}))

```

Figure 60: Best model selection using low RMSE criterion

The figure illustrates the automatic process of choosing the best model by a clear rule of evaluation. The code will compare the RMSE values of all the trained models and select the most optimal one, since the number depicts the similarity between the predictions and the real values. The use of the model ensures that the selection exercise is not done manually thus ensuring that the selection process is fair and reliable.

Results:

	Selection Rule	Best Model
0	Lowest RMSE	Linear Regression

Figure 61: Result of best model

The results of such a selection process are also shown in the figure. It unveils that the Linear Regression model was selected as the best of the analyzed models since it has the lowest RMSE. This proves that Linear Regression provides the best predictions to the COVID-19 data and is hence integrated as the final model to predict the confirmed cases.

❖ Feature Importance calculation code for best model

```

#Feature Importance
if best_model_name == "Linear Regression":
    importance_df = pd.DataFrame({
        "Feature": features,
        "Coefficient": best_model.coef_
    })
else:
    importance_df = pd.DataFrame({
        "Feature": features,
        "Importance": best_model.feature_importances_
    })

display(importance_df)

plt.figure(figsize=(10, 6))
plt.barh(importance_df["Feature"], importance_df.iloc[:, 1])
plt.gca().invert_yaxis()
plt.title("Feature Importance")
plt.show()

```

Figure 62: Feature importance

The figure illustrates the code that computes and visualizes the importance of the feature to the selected best model. Since Linear Regression is the best of all, we can extract the coefficients, which are the indications of how the features affect the prediction. The code then tabulates these coefficients so that they can be interpreted easily. It then creates a horizontal bar chart to enable us to easily compare the contribution of each feature in the prediction of confirmed COVID-19 cases.

Results:

The figure shows the feature importance in a graph and a table. The influence of country confirmed is the largest to the predicted cases, then rolling averages, and lag based features are the ones reflecting the temporal trends. Recoveries, country deaths, and growth rate comparatively have little effect. These results indicate that the most powerful predictors are historical confirmed cases and aggregate data at the country level, which provides information on the key forces of COVID-19 predictions.

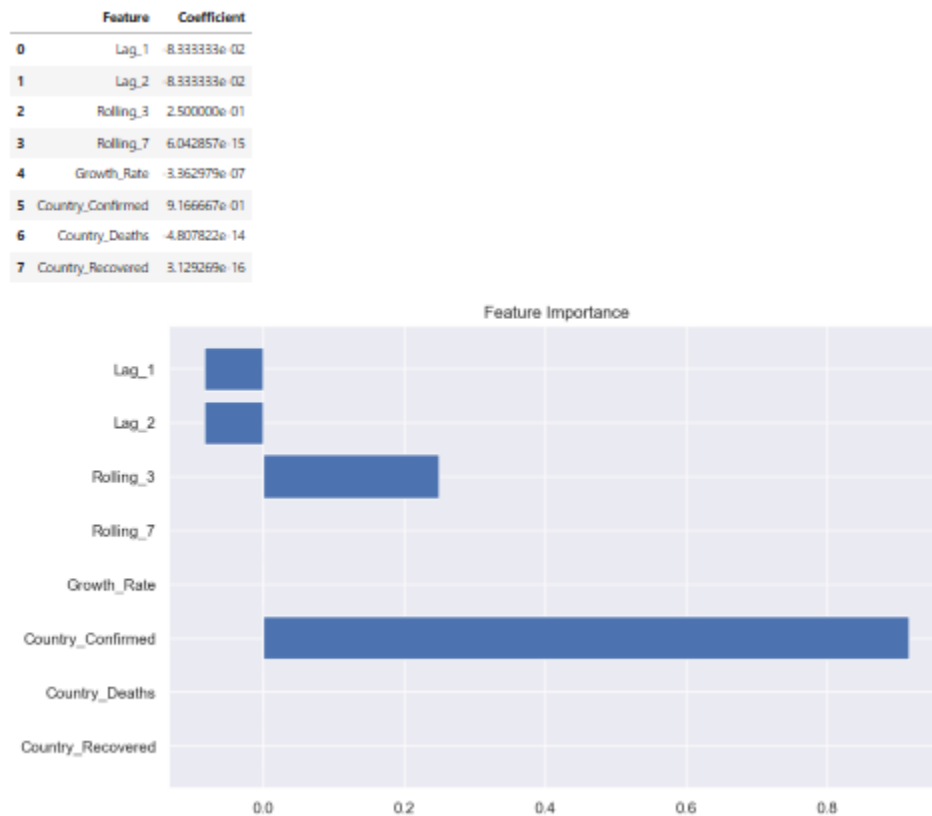


Figure 63:Feature importance Results

❖ Final Prediction Output

```
#Final Prediction Output
merged_df["Predicted_Confirmed"] = best_model.predict(X)
display(merged_df[["Date", "Confirmed", "Predicted_Confirmed"]].head())
```

Figure 64:Final Prediction

The figure describes how the final prediction output was generated by using the code. Linear Regression is the best model chosen to be used on the entire set of features to predict the confirmed cases of COVID-19. The results of the predictions are put in a new column called PredictedConfirmed and the columns chosen are shown so that they might be directly compared to the real and predicted results.

Results:

The figure describes the prediction results provided in tabular form. It shows the date-wise actual confirmed cases and the estimated confirmed cases by the model. The fact that the actual and predicted values are very close shows that the model has memorized the underlying trend which makes it equally predictive and validates the applicability of Linear Regression in case forecasting of COVID-19.

	Date	Confirmed	Predicted_Confirmed
0	2020-01-28	5580	5580.0
1	2020-01-29	6169	6169.0
2	2020-01-30	8237	8237.0
3	2020-01-31	9927	9927.0
4	2020-02-01	12038	12038.0

Figure 65: Output of Final prediction

❖ Top 10 predicted of COVID-19 Case

```
#Top-10 Prediction Visualization
top_10 = merged_df.sort_values(
    by="Predicted_Confirmed", ascending=False
).head(10)

display(top_10[["Date", "Predicted_Confirmed"]])

plt.figure(figsize=(12, 6))
plt.bar(range(1, 11), top_10["Predicted_Confirmed"])
plt.title("Top 10 Predicted COVID-19 Case Dates")
plt.xlabel("Rank")
plt.ylabel("Predicted Cases")
plt.show()
```

Figure 66: Top 10 predicted of COVID-19 Case

The figure describes the code to identify and visualize the top 10 dates with the highest number of predicted COVID-19 confirmed cases. The data is ranked descending by the values of Predicted Confirmed, and first ten are taken. These dates together with their respective case counts that are predicted are shown in tabular format and a bar chart is created ranking the top 10 dates so that the maximum predicted case values can be clearly compared.

Results:

	Date	Predicted_Confirmed
809	2022-04-16	504155459.0
808	2022-04-15	503606396.0
807	2022-04-14	502892186.0
806	2022-04-13	501920234.0
805	2022-04-12	500880363.0
804	2022-04-11	499843502.0
803	2022-04-10	498868313.0
802	2022-04-09	498299996.0
801	2022-04-08	497531675.0
800	2022-04-07	496338419.0

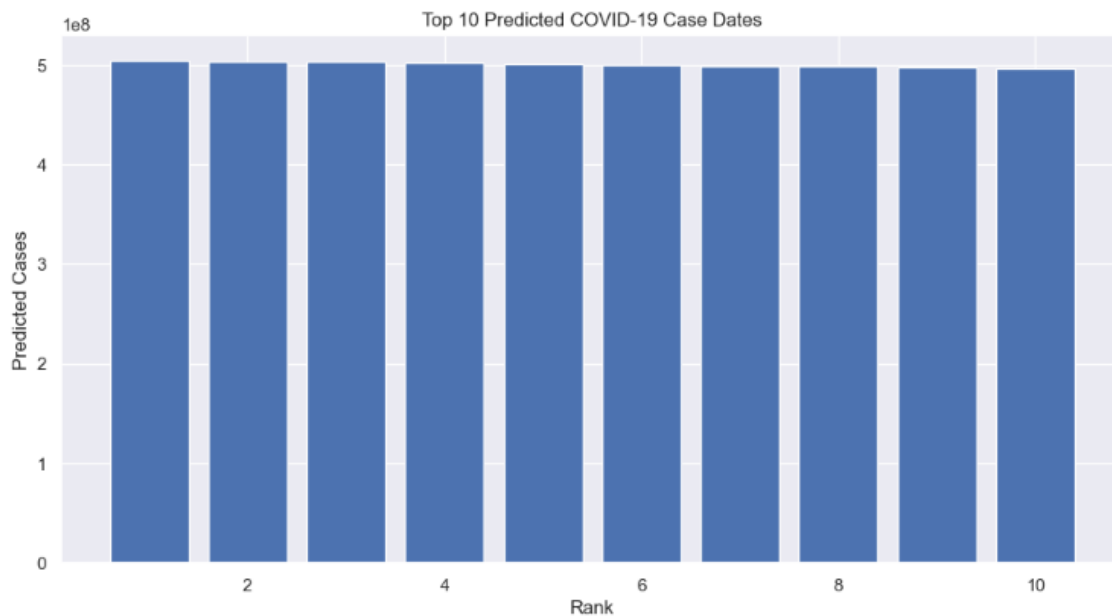


Figure 67: Visualization Output

The second figure describes the output of this visualization. It indicates the 10 most likely dates with most number of predicted COVID-19 cases and each bar indicates a ranked date with most predicted confirmed cases. The chart shows that the number of predicted cases is the highest at the later stages of the pandemic and the numbers are over 500 million, which shows the highest spreads of the virus globally. This visualization is useful to see the most important dates in terms of the severity of cases of COVID-19 with regard to prediction.

Limitations of the Proposed Model

Although the chosen Linear Regression model has been performing well, there are several shortcomings in this research. The model is particularly dependent on past trends and the assumption that future case growth will follow the same pattern in the COVID-19 scenario, which might not be accurate in instances of sudden policy set changes, the introduction of vaccinations, and the emergence of new variants. Also, there were other external factors that were not considered like the population movement, government interventions and the capabilities of the health care that were not included because the data was limited. These could affect the number of real-life cases that occur and would raise the accuracy of prediction in the event that it is included.

❖ Python Next day COVID-19 Case Forecasting with ML Models Using machine learning models.

```

1]: import numpy as np
import pandas as pd

def predict_future():
    print("--- User Input for Future Prediction ---")
    print("Please provide the required data to predict the Confirmed Cases for the next day (T).")

    try:
        # 1. Historical Data
        print("\n1. Historical Data")
        days_input = input(
            "Enter Confirmed Cases for the last 7 days (Day T-7 to Day T-1), separated by commas:\n"
            "Example: 1000, 1100, 1200, 1300, 1400, 1500, 1600\n> "
        )

        last_7_days = [float(x.strip()) for x in days_input.split(",")]

        if len(last_7_days) != 7:
            print(f"Error: Expected 7 values, got {len(last_7_days)}.")
            return

        # Feature Engineering
        lag_1 = last_7_days[-1]
        lag_2 = last_7_days[-2]
        rolling_3 = np.mean(last_7_days[-3:])
        rolling_7 = np.mean(last_7_days)
        growth_rate = (lag_1 - lag_2) / lag_2 if lag_2 != 0 else 0

        # 2. Current Totals
        print("\n2. Current Totals")
        country_confirmed = float(
            input(f"Enter Total Confirmed Cases so far (press Enter to use {lag_1}): ") or lag_1
        )
        country_deaths = float(input("Enter Total Deaths so far: "))
        country_recovered = float(input("Enter Total Recovered so far: "))

        # Prepare input features
        input_features = pd.DataFrame([
            'Lag_1': lag_1,
            'Lag_2': lag_2,
            'Rolling_3': rolling_3,
            'Rolling_7': rolling_7,
            'Growth_Rate': growth_rate,
            'Country_Confirmed': country_confirmed,
            'Country_Deaths': country_deaths,
            'Country_Recovered': country_recovered
        ])

        print("\nPrepared Features:")
        display(input_features)

        # Predictions
        print("\n--- Prediction Results ---")

        if 'lr_model' in globals():
            lr_pred = lr_model.predict(input_features)[0]
            print(f"Linear Regression Prediction: {int(lr_pred)} cases")
        else:
            print("Linear Regression model not found.")

        if 'rf_model' in globals():
            rf_pred = rf_model.predict(input_features)[0]
            print(f"Random Forest Prediction: {int(rf_pred)} cases")
        else:
            print("Random Forest model not found.")

    except ValueError:
        print("Invalid input. Please enter numeric values only.")
    except Exception as e:
        print(f"An error occurred: {e}")

# Run the prediction tool
predict_future()

```

Figure 68: Future prediction of COVID-19 using user input.

The given figure demonstrates a basic Python code that can forecast the number of confirmed cases of COVID-19 the next day with the help of machine learning. The user then inputs the confirmed case numbers in the past seven days, which aids in making the system to realize the current trends. Based on this information, the program builds useful values such as previous day cases, the average cases over a few days and the growth rate to represent the trend of change (increase or decrease) better. It also requests total confirmed cases, deaths and recoveries to provide more data. The information is then presented to trained models (Linear Regression and Random Forest) to accomplish the prediction. In general, this role demonstrates that machine learning can be applied in a convenient and simple manner to forecast the short-term trends of COVID-19 with the necessary data input provided by users.

Results:

```

--- User Input for Future Prediction ---
Please provide the required data to predict the Confirmed Cases for the next day (T).

1. Historical Data
Enter Confirmed Cases for the last 7 days (Day T-7 to Day T-1), separated by commas:
Example: 1000, 1100, 1200, 1300, 1400, 1500, 1600
> 1000, 1100, 1200, 1300, 1400, 1500, 1600

2. Current Totals
Enter Total Confirmed Cases so far (press Enter to use 1600.0): 4545
Enter Total Deaths so far: 4545
Enter Total Recovered so far: 454

Prepared Features:

```

	Lag_1	Lag_2	Rolling_3	Rolling_7	Growth_Rate	Country_Confirmed	Country_Deaths	Country_Recovered
0	1600.0	1500.0	1500.0	1300.0	0.066667	4545.0	4545.0	454.0

```

--- Prediction Results ---
Linear Regression Prediction: 4282 cases
Random Forest Prediction: 28218 cases

```

Figure 69: Output of User data input

This output will confirm that the prediction system is functioning properly between user inputs and final prediction. First, the user is asked to provide confirmed COVID-19 cases numbers during the past 7 days, as a result of which the system recognizes the current pattern of the disease. Through this data the program is able to automatically calculate key features including: The number of cases in the past day (Lag-1), the number of cases in the past day before that (Lag-2), the average number of cases in the past 3 and 7 days, and the growth rate which gives the rate at which the cases are increasing. Real-life

context is also provided to the user by showing up-to-date confirmed cases, deaths, and recoveries. The results of these calculations are properly presented in the table of prepared features, which confirms that raw input is properly converted to meaningful data to be used in prediction. Lastly, these features are used to predict the cases of the next day that are confirmed using the trained Linear Regression and Random Forest models. The difference in the values of prediction shows that various machine learning models interpret patterns in different ways which justifies the application of several models to compare and reinforces the dependability of the system.

❖ **Model Selection and Justification**

Two machine learning models are applied in this project, including Linear Regression and Random Forest Regression to enhance the reliability in predictions and to compare performance. Linear Regression is easy to use, quick and easily interpretable hence it is appropriate to determine general tendencies in the increase of COVID-19 confirmed cases. It assists in the linear relationship of the change in the number of cases with time. Random Forest Regression, in its turn, can operate with complex and non-linear trends and incorporating several decision trees. Random Forest can make stronger and more adaptable predictions since the trends of COVID-19 can change because of a number of influencing variables. The fact that both models are used shows that critical thinking is used as well as the ability to compare the performance of the models and enhance the overall credibility of the prediction system.

Table 2: Model Selection and justification Table

Criteria	Linear Regression	Random Forest Regression
Model Type	Linear, parametric	Ensemble, non-linear
Interpretability	High – easy to understand coefficients	Low – complex ensemble of trees
Ability to Capture Trends	Very effective for linear and cumulative trends	Effective for complex, non-linear patterns
MAE	Very low	High
RMSE	Very low	High
MAPE	Very low	High
Training Time	Very fast	Computationally expensive
Generalization on Time-Series Data	Strong	Weaker for this dataset
Suitability for COVID-19 Data	Highly suitable	Less suitable
Final Selection	Selected	Not Selected

4. Conclusion

4.1 Work done Analysis

The project involved the use of supervised machine learning methods to forecast the history of COVID-19 cases worldwide with publicly accessible data such as worldwide-aggregate.csv and countries-aggregate.csv. By having an in-depth examination of the daily confirmed cases, recoveries, deaths and growth rate, the study indicated that past pandemic data could be well utilized in predicting the future trend in the short term. Lag values, rolling averages, and percentages growth rate features are examples of time-series data preprocessing and feature engineering methods that greatly enhanced the quality of the input data, hitting noise reduction and real world temporal dependencies.

The Linear Regression model and the Random Forest Regression model have been compared and assessed by applying the standard evaluation indicators, such as MAE, RMSE, and MAPE. The findings revealed that Linear Regression performed more effectively in prediction, which means that there is a high cumulative and linear trend in the development of COVID-19 cases. Random Forest Regression can also be used to model non-linear patterns but it proved to be weaker at generalizing when using this particular time-series data. These results indicate the necessity of choosing models of machine learning according to the specifics of the data and not the complexity of models only. In general, the created system is a population-level trend-predicting model, not an individual diagnostic model; hence, it can be applied to the big-scale decision-making in the field of public health.

Reflective Insight:

The presented research provides a valuable note about artificial intelligence and machine learning: a more complicated model does not necessarily lead to a higher predictive accuracy. Although ensemble algorithms like the Random Forest Regression can be strong and can observe the non-linear trends, the performance of the algorithm will highly rely on the behaviour of data. In the present case, the original data of COVID-19 time-series has a significant linear trend, which means that the

Linear Regression model is superior to more sophisticated ones. This consideration makes it clear that great caution should be used when choosing an algorithm depending on the characteristics of the data instead of the belief that complex models will provide better results.

4.2 Real-World Impact and Significance

The proposed AI-based forecasting system proves to be practically valuable in assisting the decision-making process of the population under the conditions of a pandemic. The system will be able to help healthcare officials and government bodies predict the occurrence of infection outbreaks by giving short-term forecasts of the COVID-19 cases trends. This allows better capacity and medical resource planning, staffing, and emergency response planning of the hospital.

Also, with the controlled machine learning, the system is able to learn trends using past information and keep on improving predictions as new information is provided. This is particularly beneficial during health emergencies like the COVID-19 pandemic that change at a high rate, with quick trends identification being able to greatly ease the burden on health services. Altogether, the solution illustrates how artificial intelligence can be practically applied to the real-life issues in the domain of global health as it provides timely and data-driven information.

4.3 Further Work

❖ Improvement of Accuracy of Prediction.

- **Additional Data:** Additional data should be incorporated in the report. Vaccination rates, government intervention policies, population density, mobility data, and demographic information are some of the other features that can be incorporated to improve the predictive performance of the model. The model would be more effective in including external factors that may influence the transmission of the virus.

- **Improved Time Series Models:**

More effective forecasting can be achieved by using more complex and long-term relationships in the data, which are better represented using state of the art time series or deep-learning techniques.

❖ Country-Specific Predictions

Localized Predictions:

To make the model more relevant in practice, it would be beneficial to extend it further to be able to make country-level or regional predictions and allow healthcare planning and policy response to be tailored to specific countries, especially those with medical resources at their disposal.

❖ Real-Time Data Updates

Dynamic Forecasting:

Real-time updates of the data used in the forecasting system would enable it to be always able to adjust to new trends and the predictions would be more responsive and reliable when it comes to dynamic monitoring of health conditions of people and decision-making.

5. References

- Anaconda, 2024. *Anaconda Documentation*. [Online]
Available at: <https://www.anaconda.com>
[Accessed 30 12 2025].
- Breiman, L., 2001. Random Forests. *Machine Learning*, Volume 45, pp. 5-32.
- Chai, T. & D. R. R., 2014. Root mean square error (RMSE) or mean absolute error (MAE)?.. *Geoscientific Model Development*, Volume 7, p. 71247–1250..
- Chimmula, V. K. R., 2020. Time series forecasting of COVID-19 transmission in Canada using LSTM networks. *Chaos, Solitons & Fractals*, Volume 135.
- Domenico Benvenuto, M. G., April 2020. Application of the ARIMA model on the COVID-2019 epidemic dataset. *Sciencedirect*, Volume 29.
- Ensheng Dong, H. D. a. L. G., 2020. *An interactive web-based dashboard to track COVID-19 in real time*, s.l.: sciencedirect.
- Google, 2020. *Google COVID-19 mobility reports*.. [Online]
Available at: <https://www.google.com/covid19/mobility>
[Accessed 16 Dec 2025].
- Hale, T. A. N. G. R. e. a., 2021. A global panel database of pandemic policies (Oxford COVID-19 Government Response Tracker. *Nature Human Behaviour*, Volume 5, pp. , 529–538.
- Harris, C. R. e. a., 2020. Array programming with NumPy. *Nature*, Volume 585, p. 357–362.
- HopkinsUniversity, J., 2020. *COVID-19 data repository*. [Online]
Available at: <https://coronavirus.jhu.edu/data>
[Accessed 15 Dec 2025].
- Hyndman, R. J. & A. G., 2018. *Forecasting: Principles and Practice*., s.l.: OTexts..
- Hyndman, R. J., 2018. *Forecasting: Principles and Practice*. OTexts.
- Kluyver, T. e. a., 2016. *Jupyter Notebooks*, s.l.: IOS Press.
- Kotsiantis, S. B. K. D. & P. P. E., 2006. Data preprocessing for supervised learning.. *International Journal of Computer Science*, Volume 1(2), pp. 111-117.
- Limao Zhang, Y. P. M. S., 2021. *Introduction to Artificial Intelligence*, s.l.: Lecture Notes in Civil Engineering.
- McKinney, 2010. *Data structures for statistical computing in Python*.. s.l., Proceedings of the 9th Python in Science Conference.
- McKinney, W., 2010. Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*, p. 51–56..

- Mohammad Ayoub Khan, M. A. K. ., A., 2022. Performance evaluation of regression models for COVID-19: A statistical and predictive perspective. *Ain Shams Engineering Journal*, 13(2).
- Montgomery, D. C. P. E. A. & V. G. G., 2012. *Introduction to Linear Regression Analysis*, s.l.: Wiley.
- Montgomery, D. P. E. a. V. G., 2012. *Introduction to Linear Regression Analysis*. 5th edn ed. s.l.:Hoboken: Wiley..
- Özen, F., 29 February 2024,. Random forest regression for prediction of Covid-19 daily cases and deaths in Turkey. *Sciencedirect*, 10(4).
- Pedregosa, F. e. a., 2011. Scikit-learn: Machine learning in Python.. *Journal of Machine Learning Research*,, Volume 12, p. 2825–2830..
- Pratiksha Tidke, S. S. P. G., 2023. *A Review on Weather Forecasting using Linear Regression*, s.l.: International Scientific Journal of Engineering and Management.
- Russell, S. & N. P., 2021. *Artificial Intelligence: A Modern Approach (4th ed.)*, s.l.: Pearson.
- Russell, S. a. N. P., 2021. *Artificial Intelligence: A Modern Approach*. 4th edn. ed. Harlow: Pearson Education..
- Shinde, G. R. K. A. B. M. P. N. D. N. C. J. & H. A. E., 2020. Forecasting models for coronavirus disease (COVID-19): A survey. *EEE Access*, Volume 8.
- Shreshth Tuli, S. T. R. T. ., S. G., 2020. Predicting the growth and trend of COVID-19 pandemic using machine learning and cloud computing. *Internet of Things*, Volume 11.
- Shrikant Tiwari, P. C. ., S. K. S., 2022. A Review of the Machine Learning Algorithms for Covid-19 Case Analysis. *National Library of Medicines*.
- Trevor Hastie, R. T. ., J. F., 2009. The Elements of Statistical Learning. In: *Data Mining, Inference, and Prediction*. Springer.. s.l.:SpringerNaturelink.
- Tuli, S. T. S. T. R. & G. S. S., 2020. Predicting the growth and trend of COVID-19 pandemic using machine learning and cloud computing.. *Internet of Things*,, Volume 11.
- Tuli, S. T. S. V. R. a. T. R., 2020. Predicting the growth and trend of COVID-19 pandemic using machine learning and cloud computing'. *Internet of Things*, p. 11.
- Waskom, M. L., 2021. Seaborn: Statistical data visualization. *Journal of Open Source Software*, Volume 6, p. 60.
- WHO, 2020. *COVID-19 situation reports*., s.l.: World Health Organization.
- Wilkinson, L. a. F., 2009. The history of the cluster heat map. *The American Statistician*, Volume 63(2), p. 179–184.
- Willmott, C. J. & M. K., 2005. Advantages of the mean absolute error over the root mean square error in assessing average model performance.. *Climate Research*, Volume 30(1), pp. 79-82.

Wu, J. T. L. K. & L. G. M., 2020. Nowcasting and forecasting the potential domestic and international spread of the 2019-nCoV outbreak originating in Wuhan, China.. *The Lancet*, Volume 395(10225), p. 689–697.

Yasminah Alali, F. H. & Y. S., 2022. *A proficient approach to forecast COVID-19 spread via optimized dynamic machine learning models*, s.l.: Scientific Reports.

Yingyu Yin a, I. A. ,. K. K. a. H. E., 2024. Advanced forecasting of COVID-19 epidemic: Leveraging ensemble models, advanced optimization, and decomposition techniques. *Computers in Biology and Medicine*, Volume 175.

Yunyun Cheng, R. C. ,. T. X. ,. X. T. ,. Y. B., 2025 May 13. *Machine Learning Techniques Applied to COVID-19 Prediction: A Systematic Literature Review*, s.l.: National Library of Medicine.